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Masi

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(54) **PERSONAL FLOATATION DEVICE**

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B63B 7/08 (2020.01)
B63B 34/52 (2020.01)
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(52) **U.S. Cl.**
CPC **B63B 34/565** (2020.02); **B63B 7/085** (2013.01); **B63B 34/52** (2020.02); **B63C 9/30** (2013.01)

(58) **Field of Classification Search**

CPC B63B 34/565; B63B 34/52; B63B 7/085
See application file for complete search history.

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Primary Examiner — S. Joseph Morano

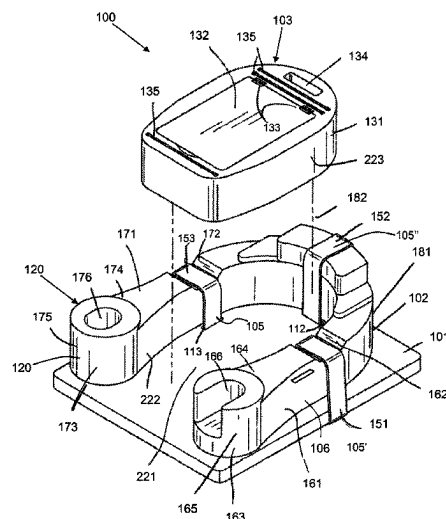
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(57) **ABSTRACT**

A personal floatation system provides a horseshoe shaped float that is buoyant and can be used in a number or positions to float a person on the water and retain a beverage. The horseshoe shaped floatation device has a left and right armrest that extend out from an arced shaped closed end and dual cup holder, on the extended end of the armrest, has a cup cylinder to retain a beverage in the vertical orientation and an end cup holder to retain a beverage in an orthogonal orientation to the cup cylinder. The floatation device may have an extending strap that couples the left and right armrests together to secure the horseshoe shaped float around a person's torso.

17 Claims, 21 Drawing Sheets



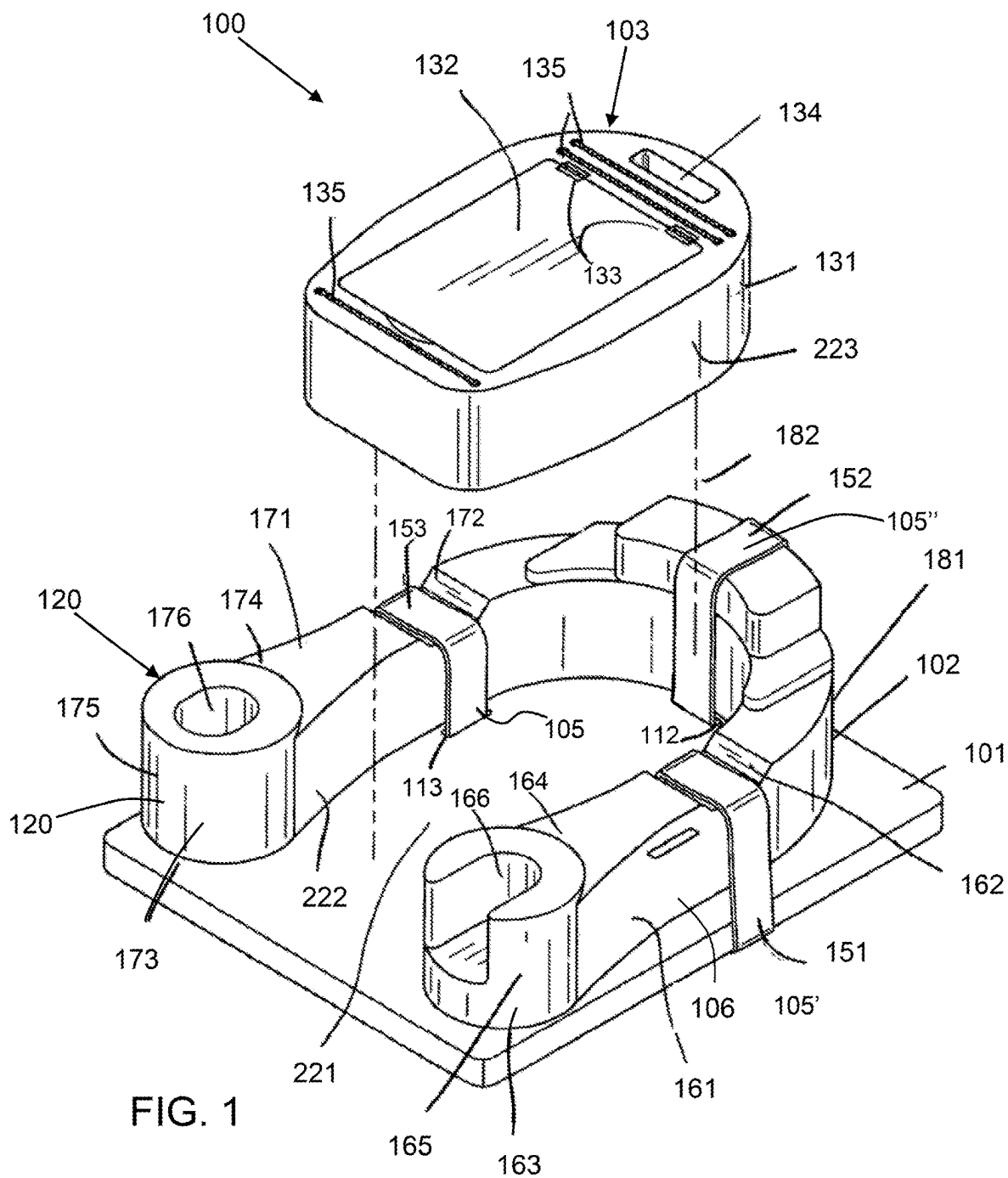
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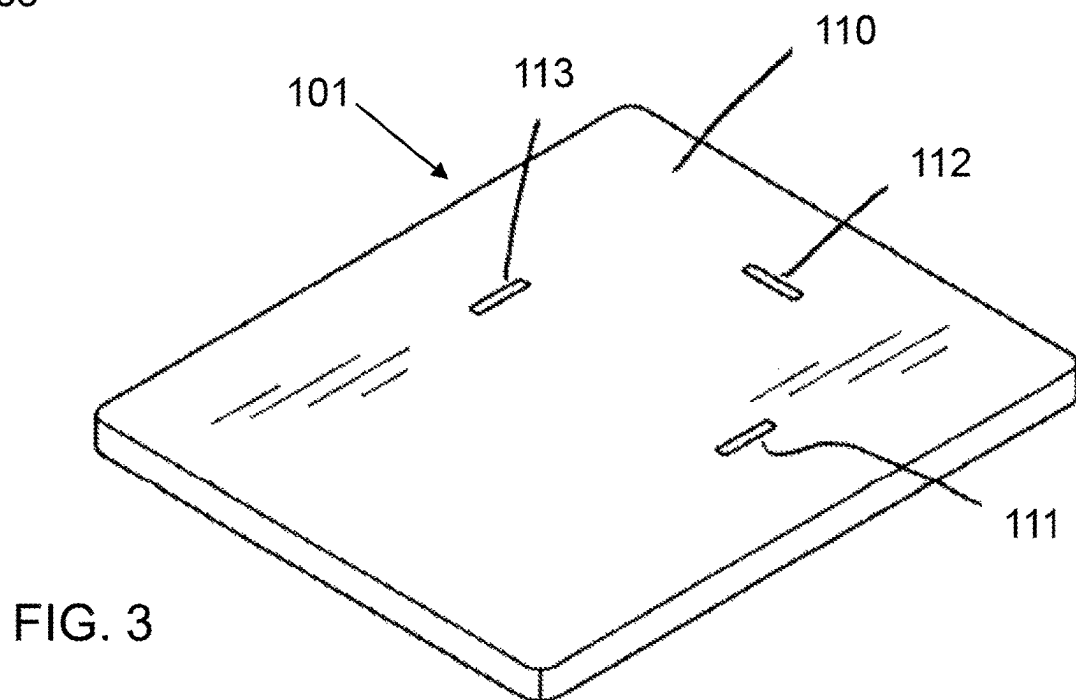
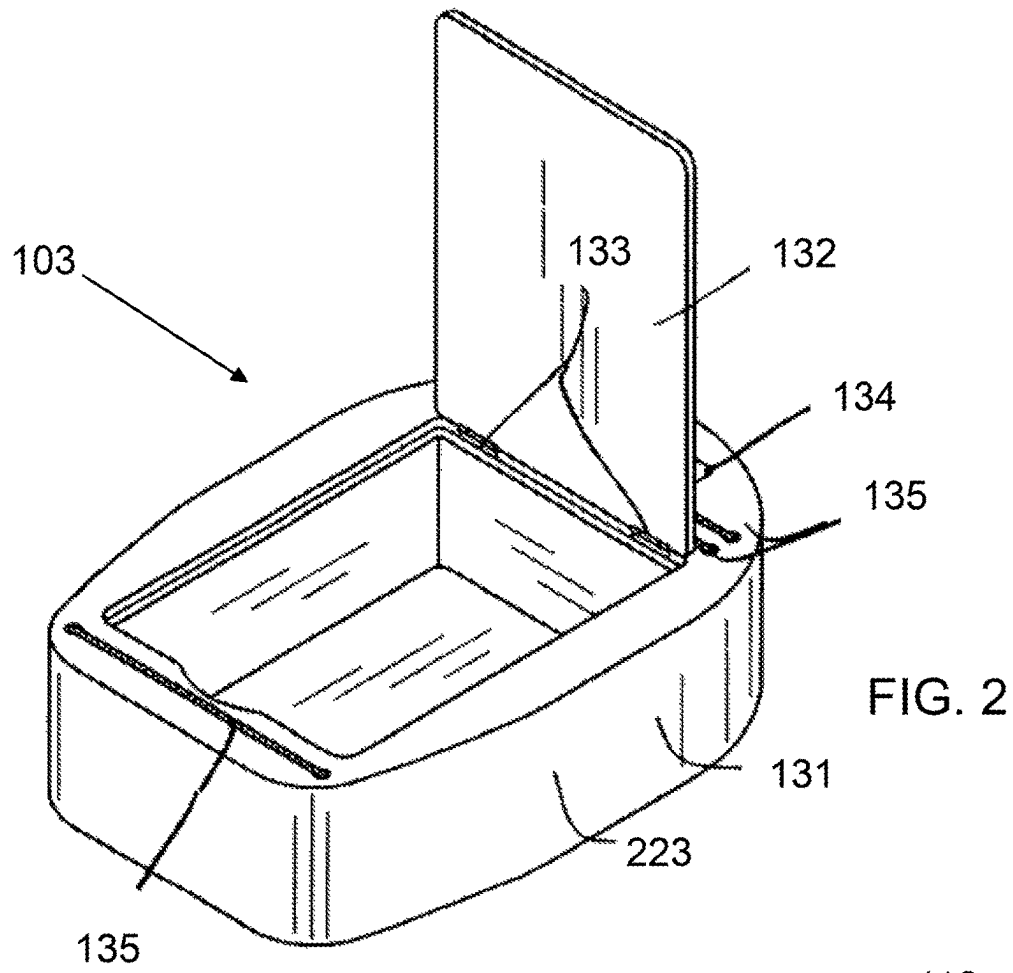
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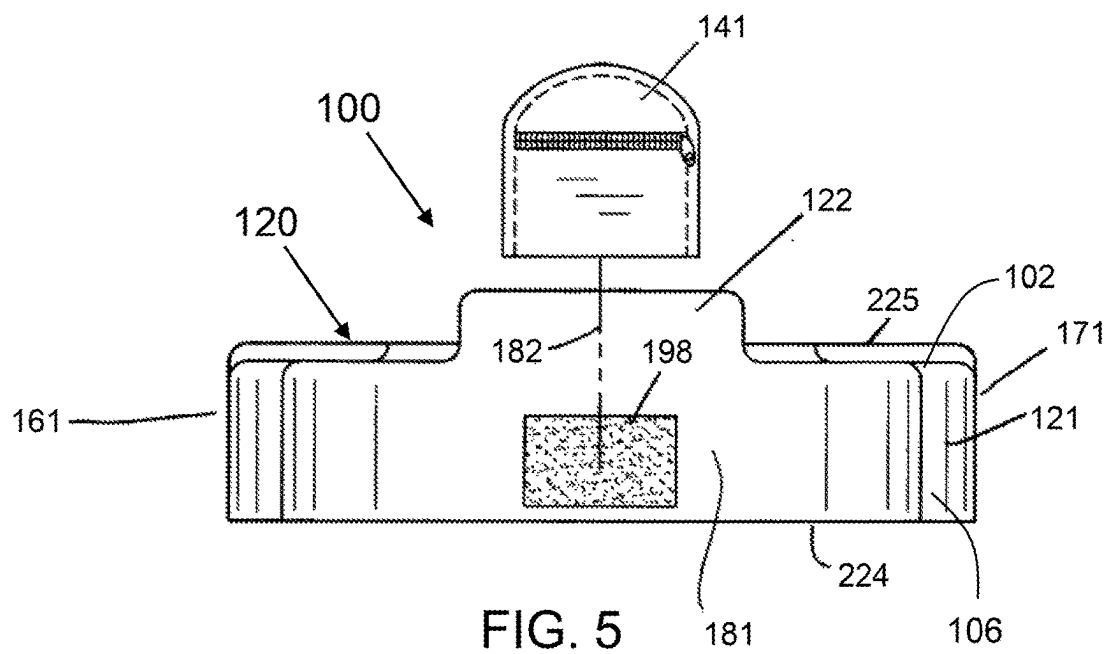
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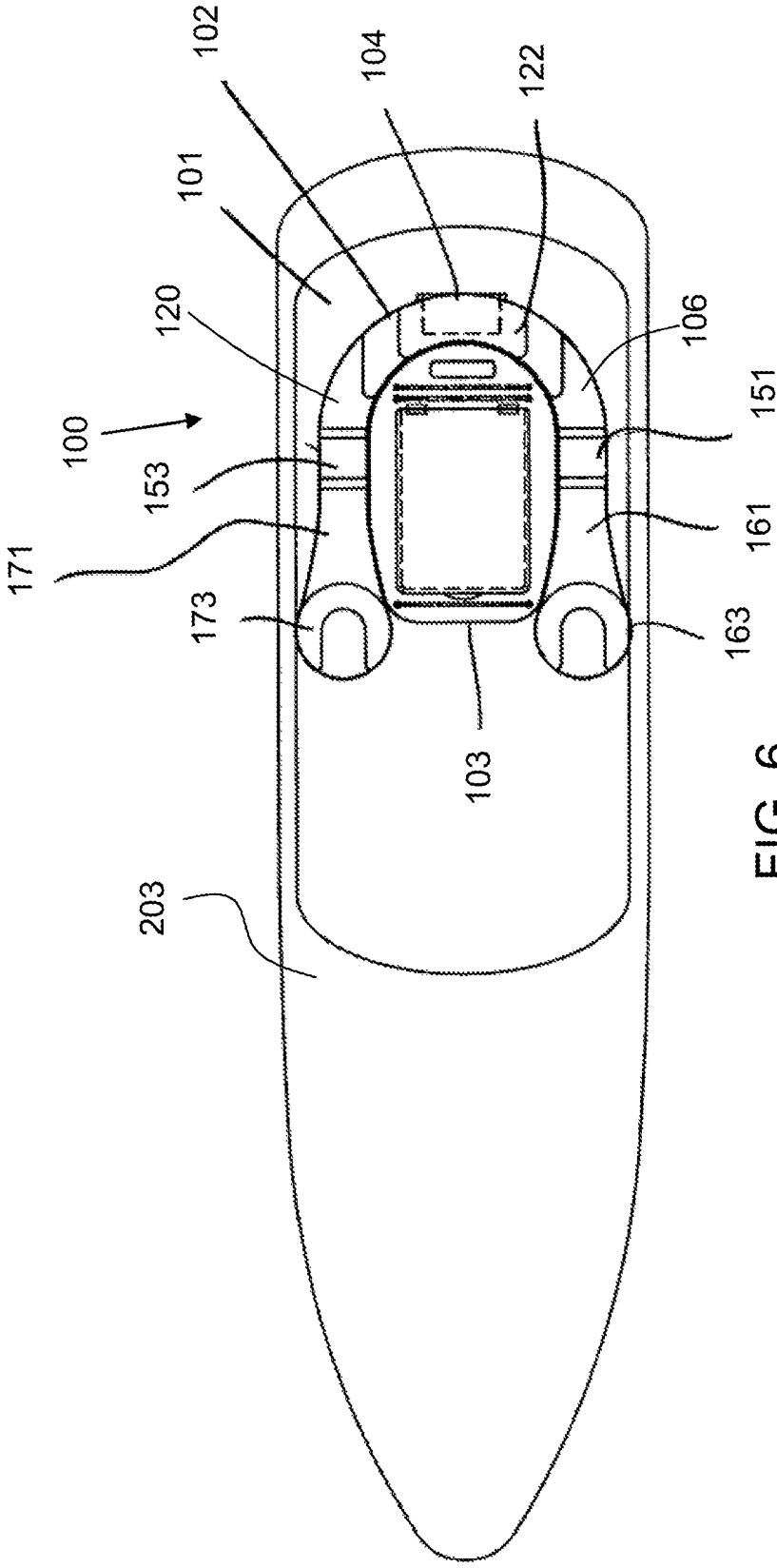


FIG. 6

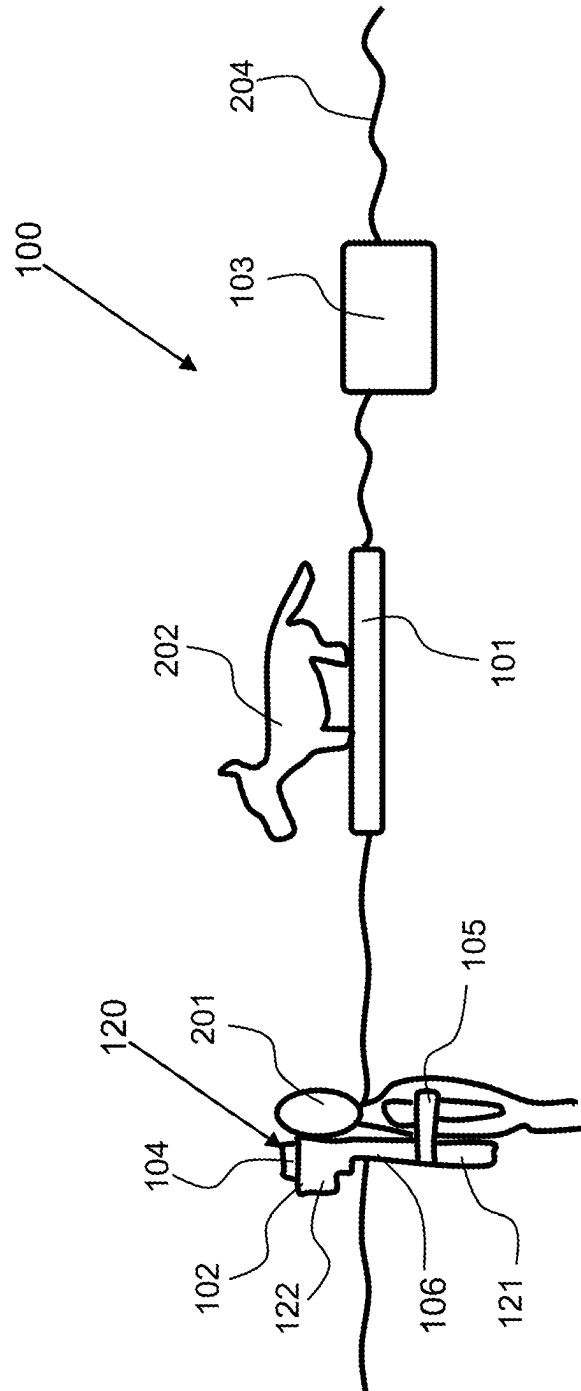


FIG. 7

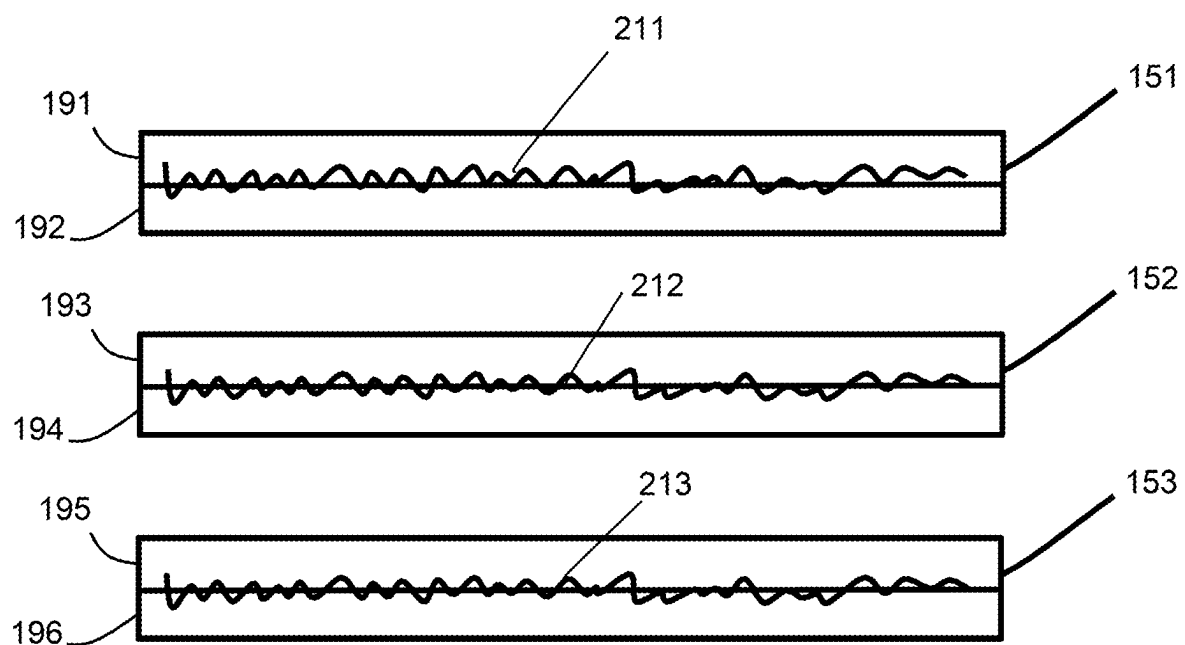


FIG. 8

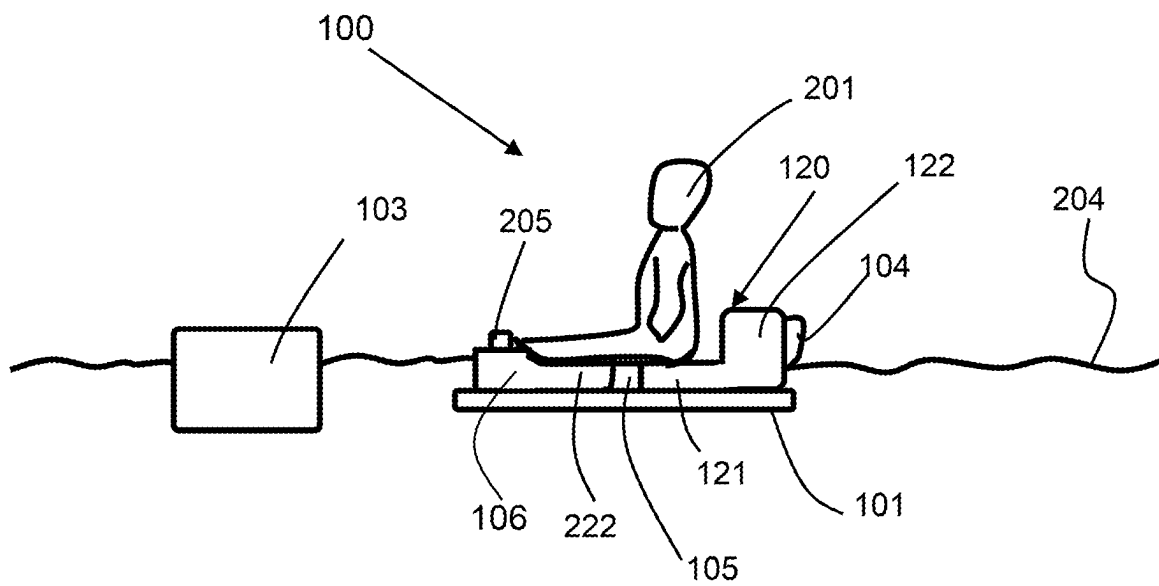


FIG. 9

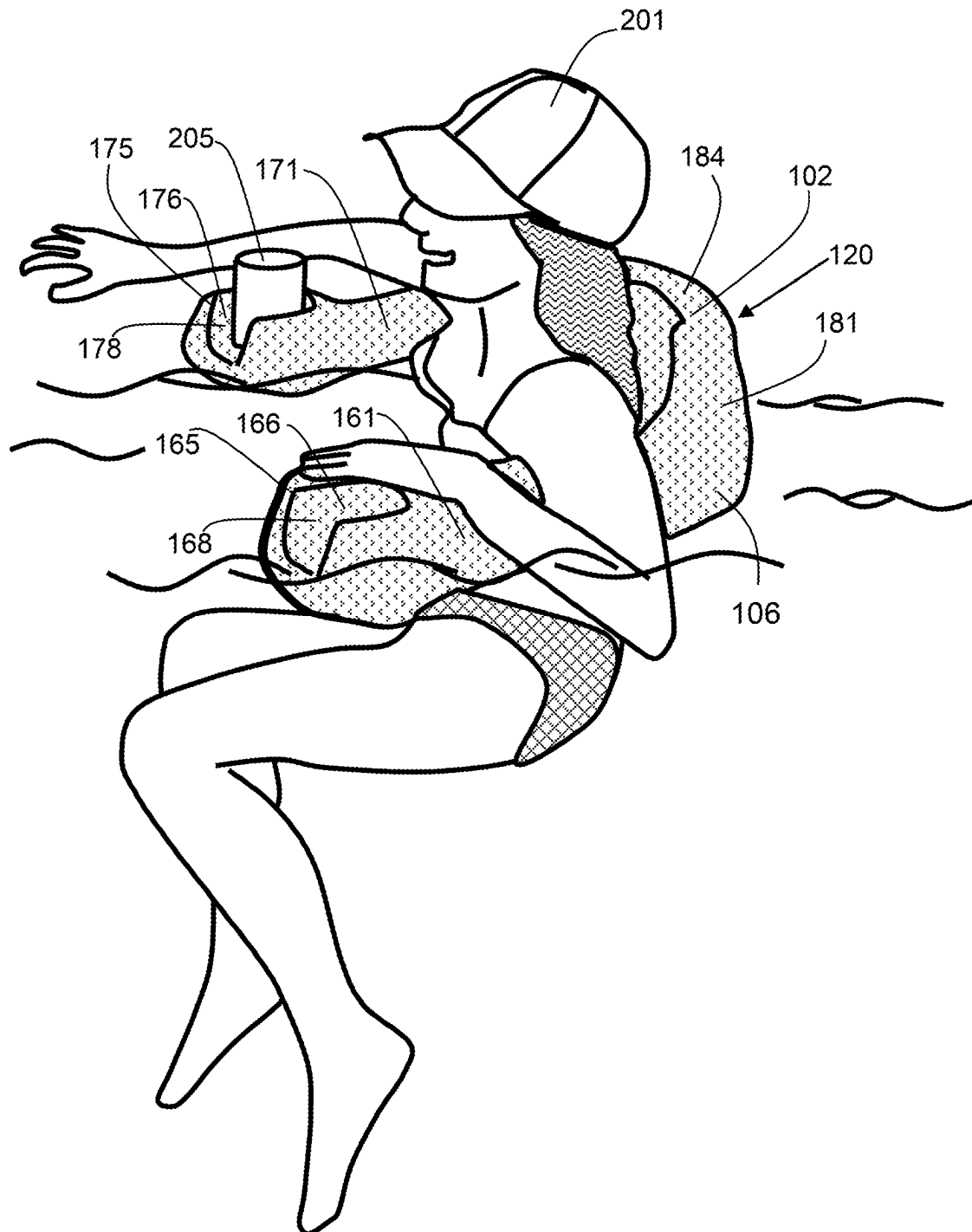


FIG. 10

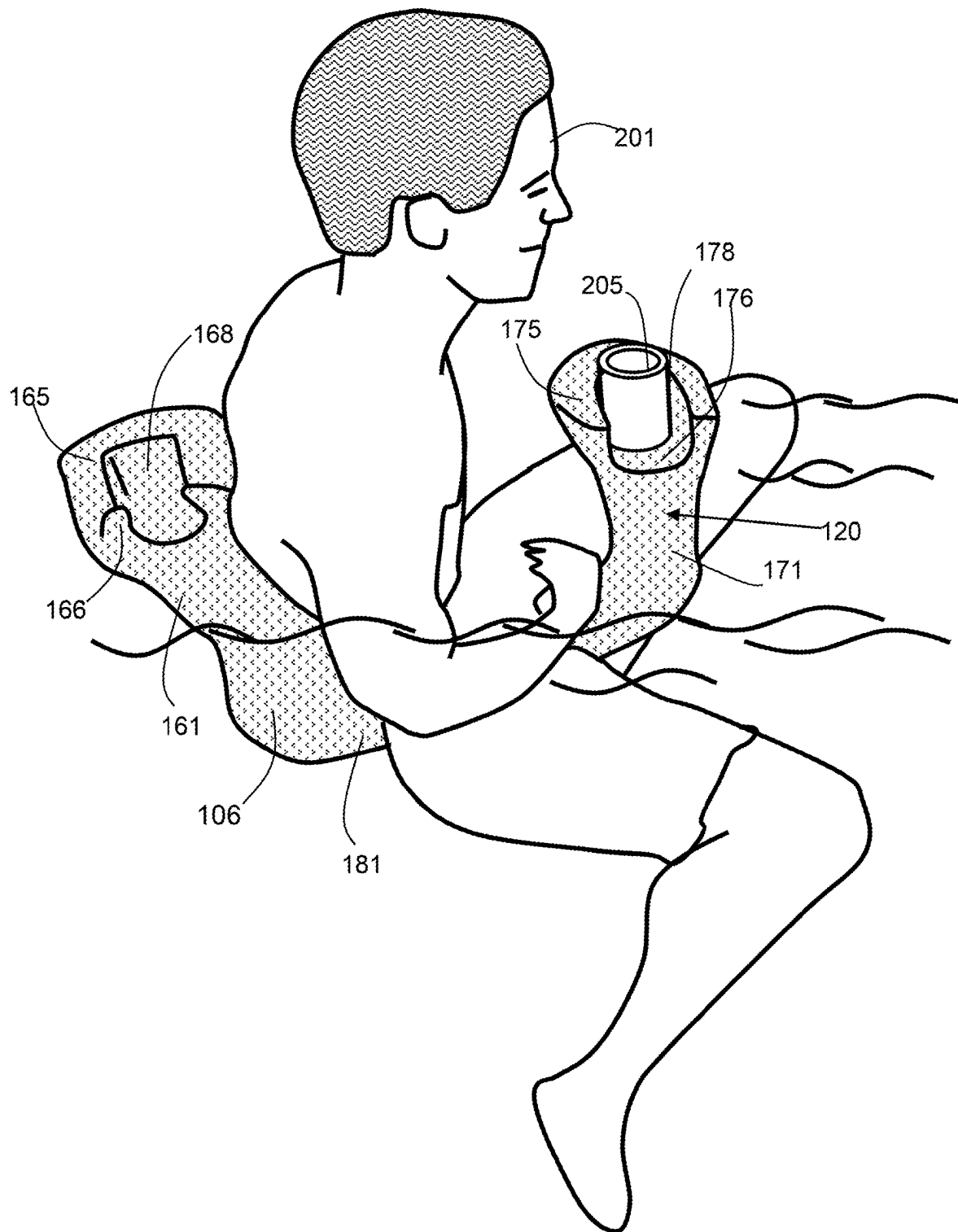


FIG. 11

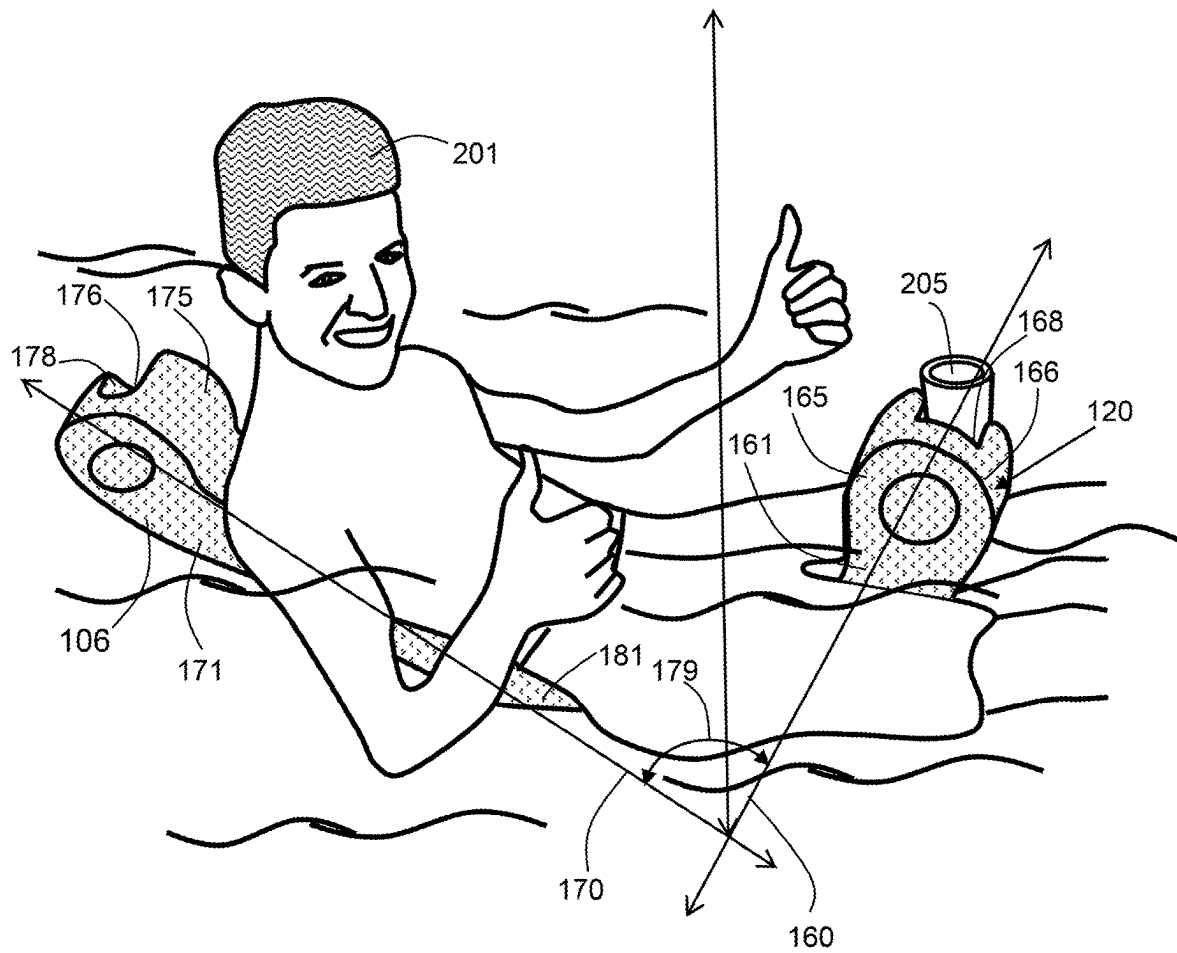


FIG. 12

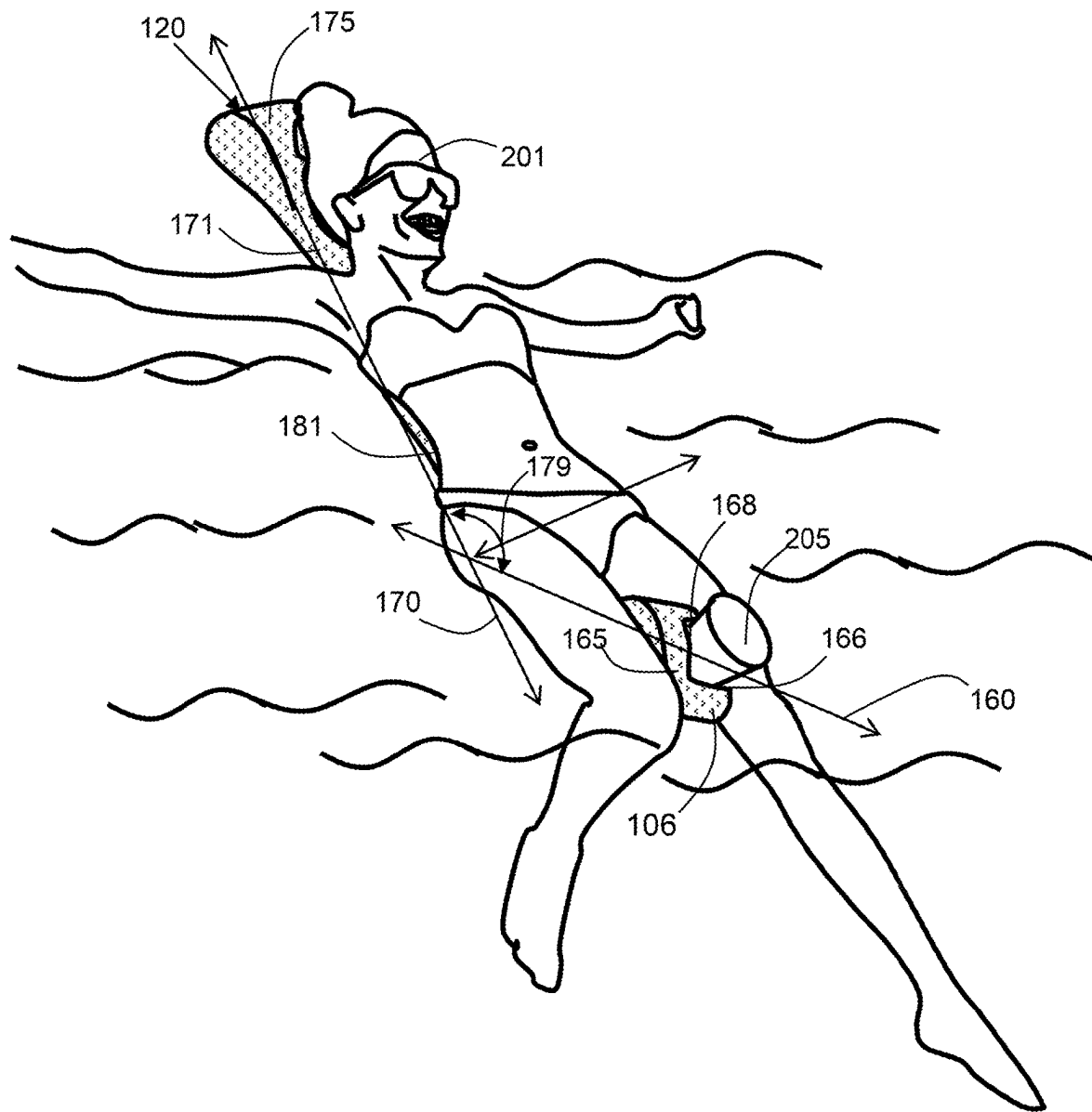


FIG. 13

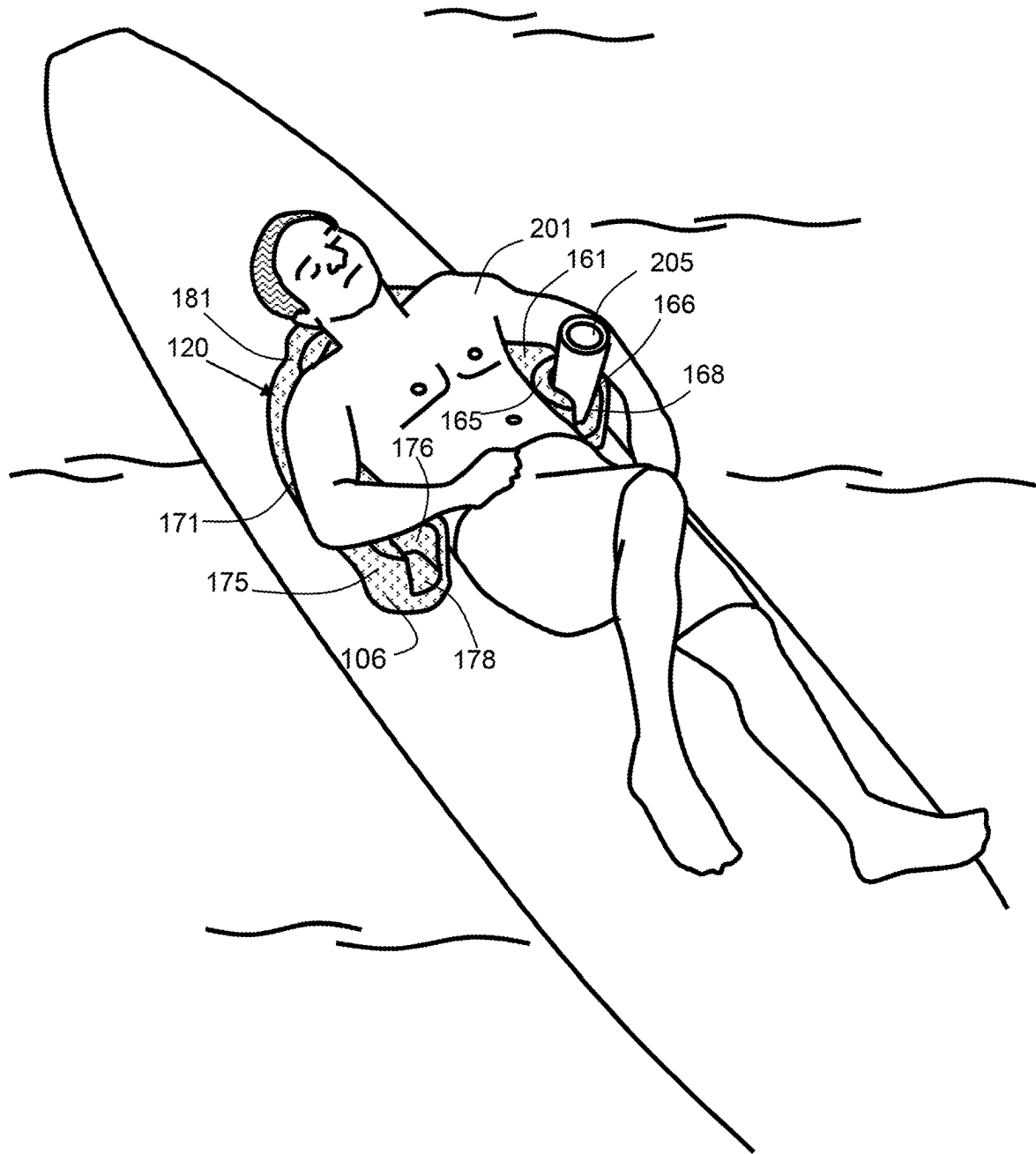


FIG. 14

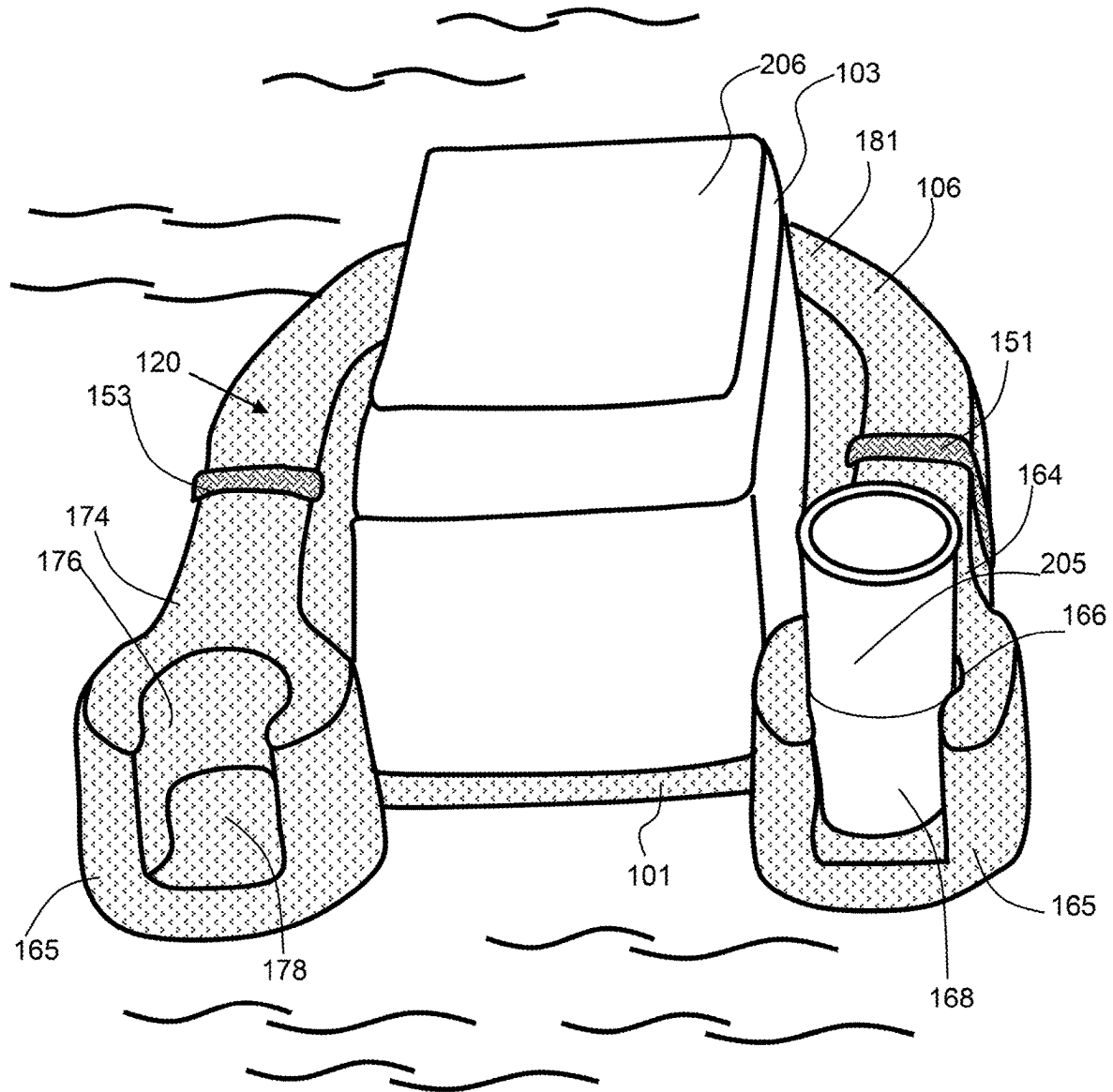


FIG. 15

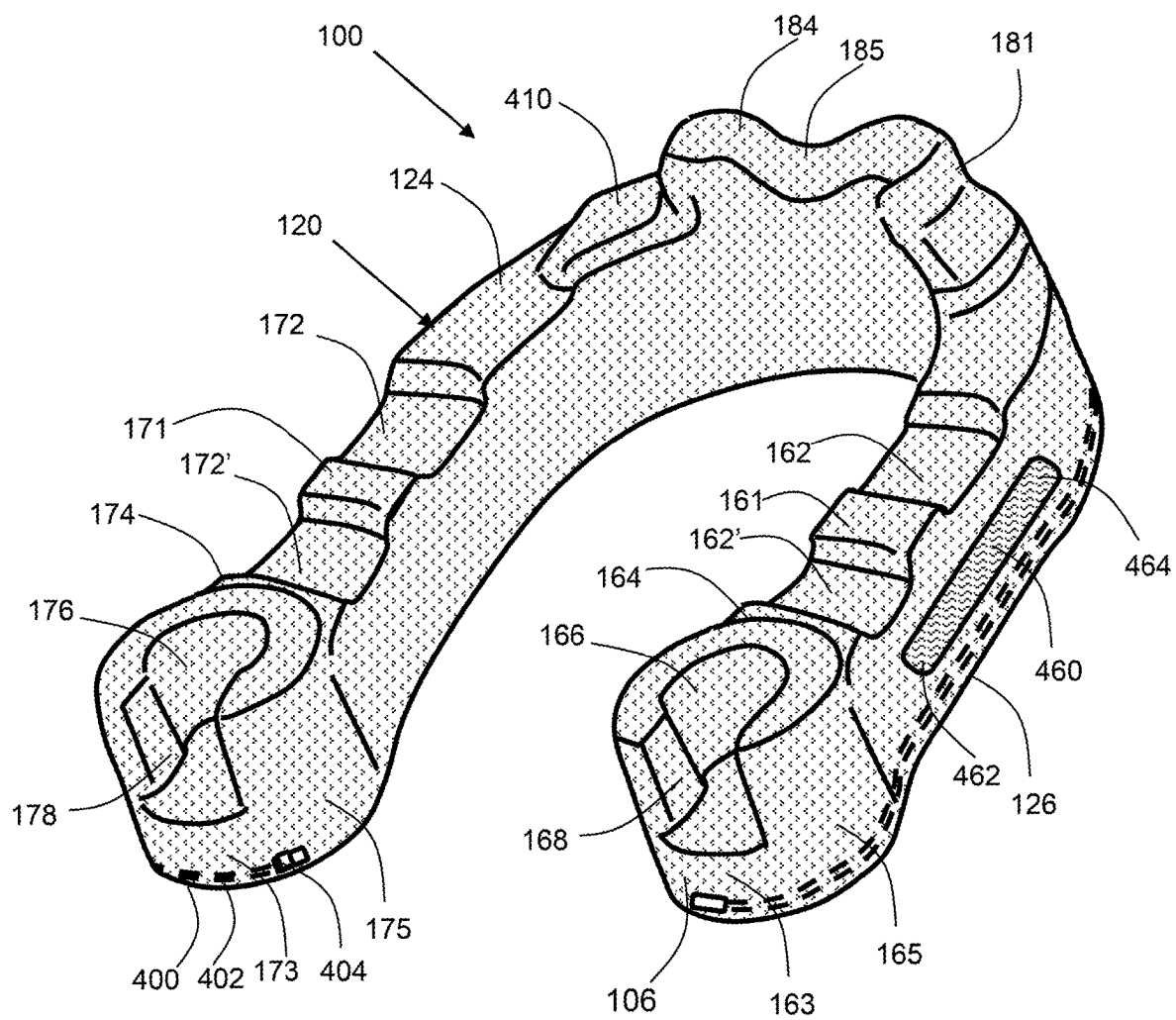
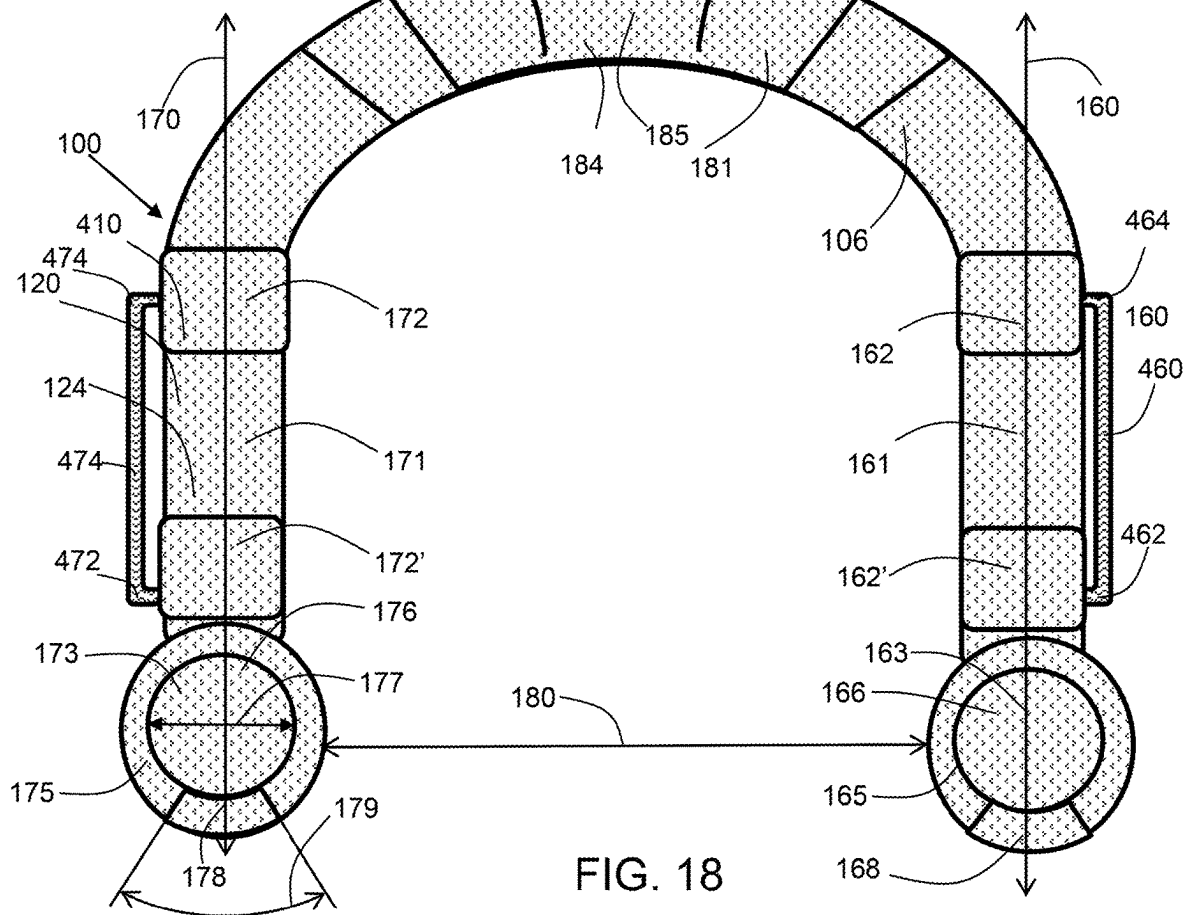
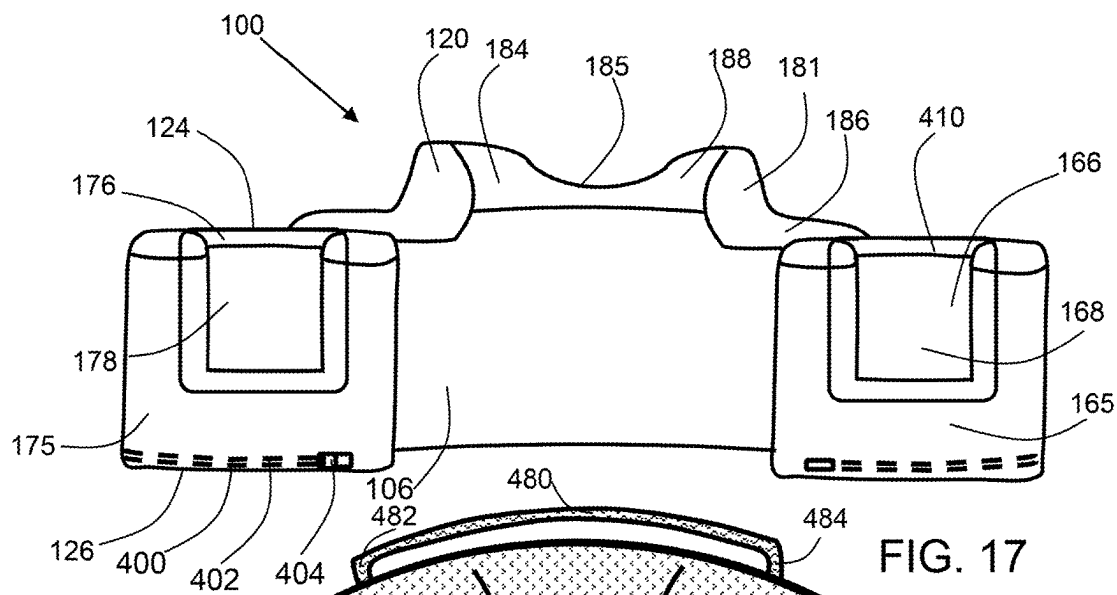


FIG. 16



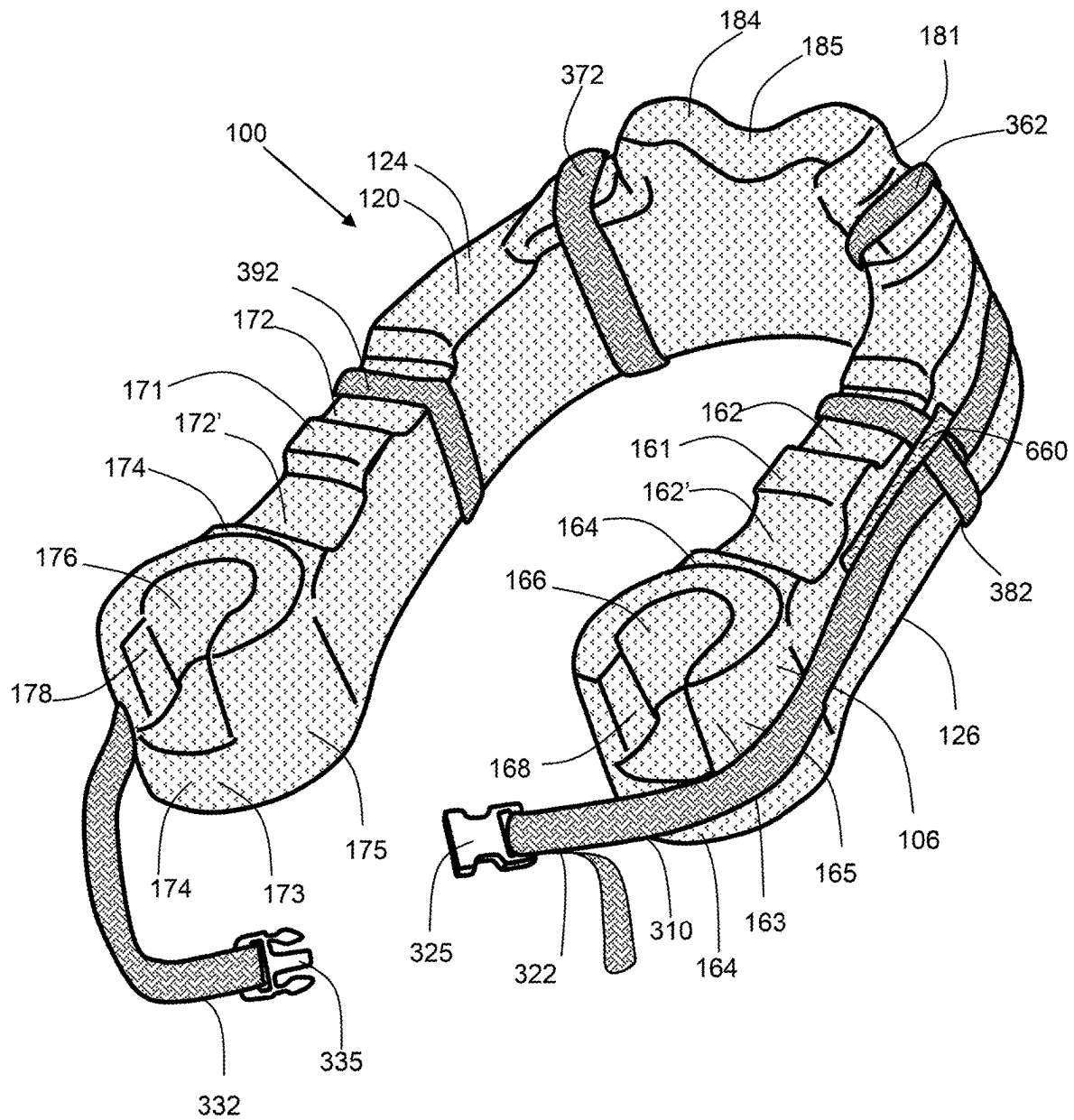


FIG. 19

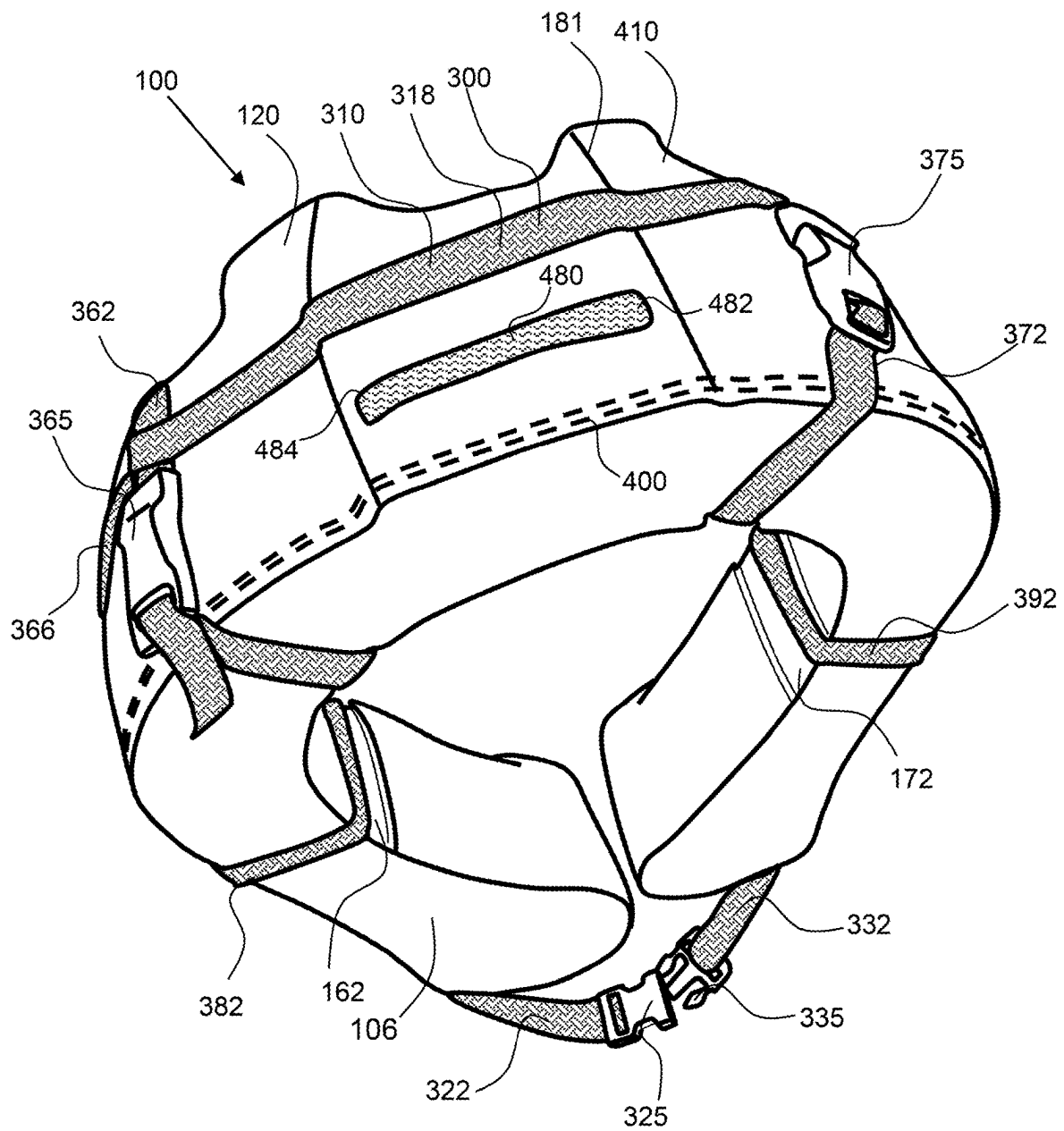


FIG. 20

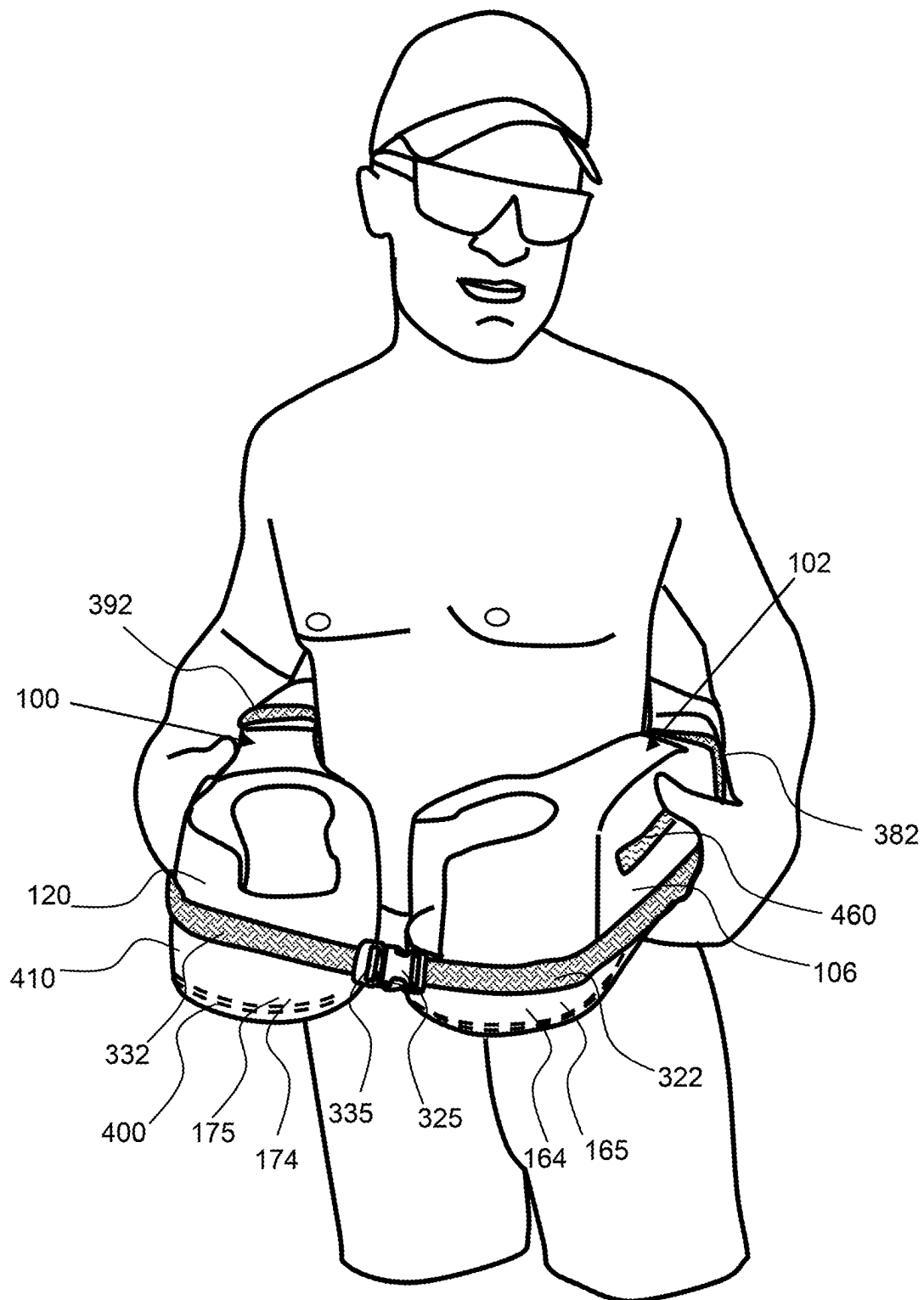


FIG. 21

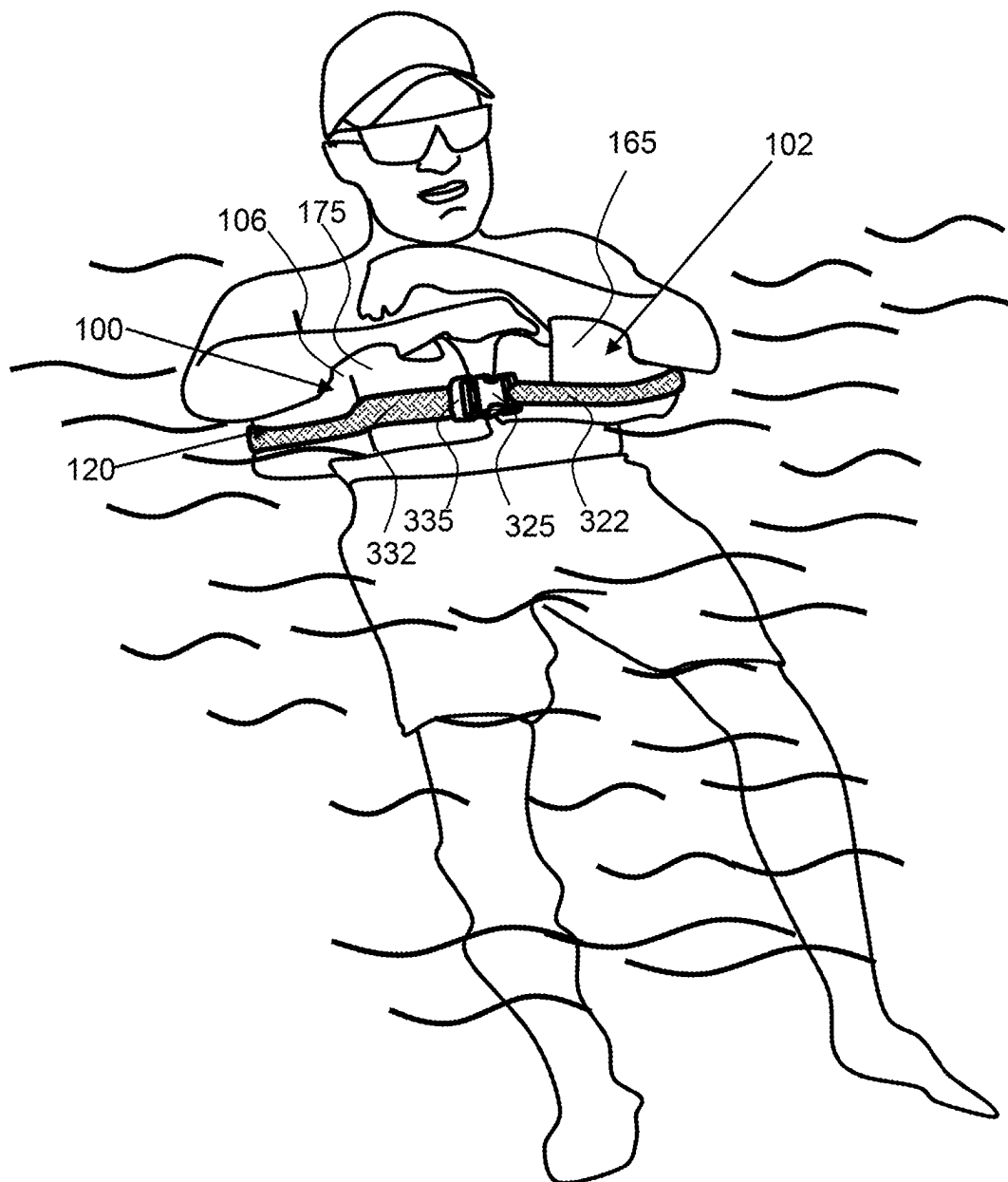


FIG. 22



FIG. 23

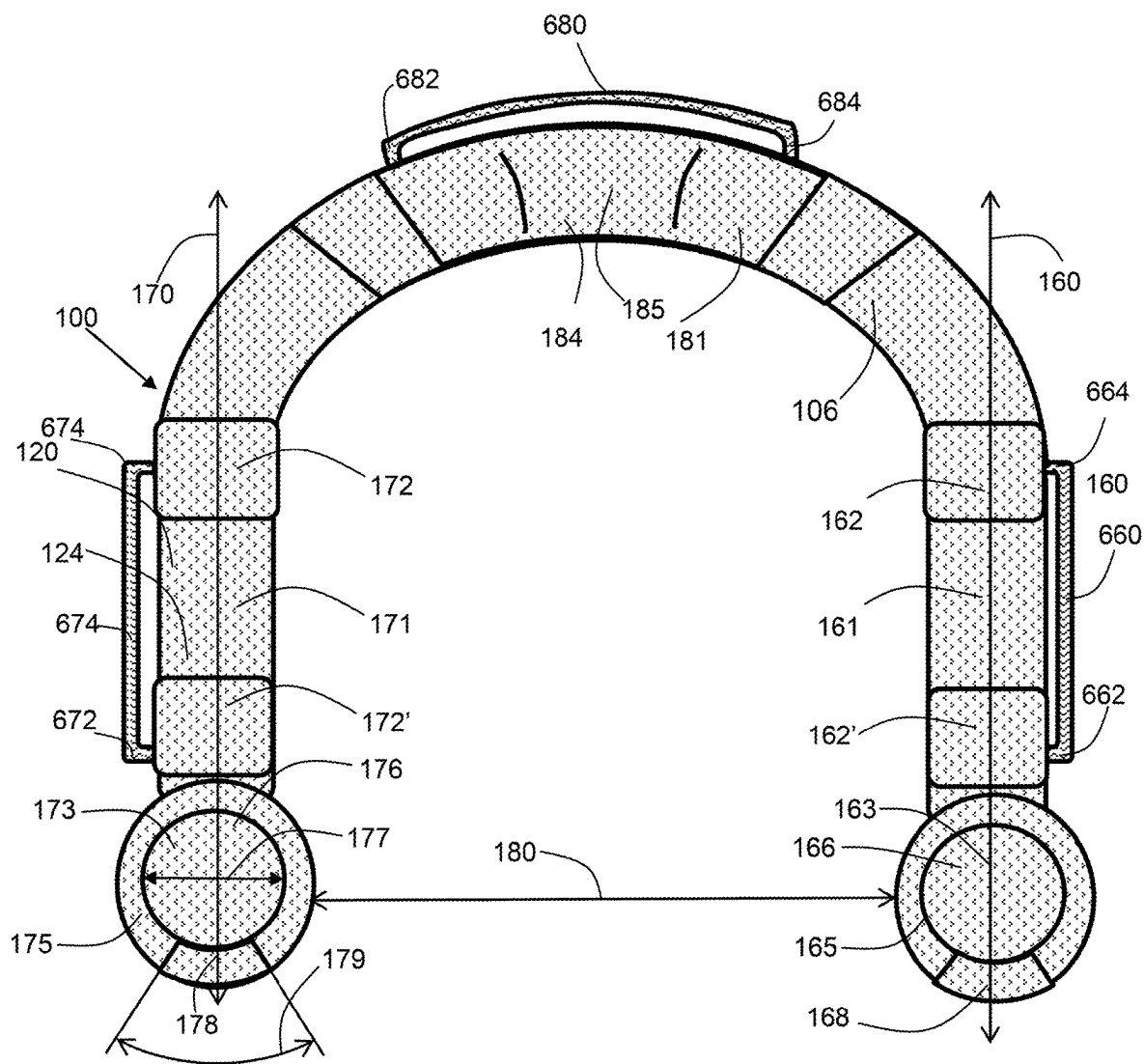


FIG. 24

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PERSONAL FLOATATION DEVICE**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a continuation in part of U.S. patent application Ser. No. 17/080,338, filed on Oct. 26, 2020 and currently pending, which is a continuation in part of U.S. patent application Ser. No. 16/200,218, filed Nov. 26, 2018 and issued as U.S. Pat. No. 10,814,943 on Oct. 27, 2020, which is a continuation in part of Ser. No. 15/728,866, filed on Oct. 10, 2017 and now issued as U.S. Pat. No. 10,219,633 on Mar. 5, 2019 and entitled Personal Floatation Device, which claims the benefit of U.S. provisional patent application 62/409,527 filed on Oct. 18, 2016, and U.S. application Ser. No. 16/200,218 claims the benefit of priority to U.S. provisional patent application No. 62/691,882, filed on Jun. 29, 2018; the entirety of each is hereby incorporated by reference herein. Floatation

BACKGROUND OF THE INVENTION**Field of the Invention**

The present invention relates to the field of transporting and ships including equipment for life-saving in water, more specifically, an adaptation of vessel parts of furnishing for lifesaving purposes for use with people and animals.

Background

Personal floatation devices (PFDs) typically have a fixed shape which limits the ways in which they can be used. There exists a need for a personal floatation device that can be used in a variety of configurations, that is flexible and has retainers for beverages.

SUMMARY OF THE INVENTION

The personal floatation device is adapted for use with a person. The personal floatation device is adapted for use with a pet. The personal floatation device (PFD) is configured for use with water. The personal floatation device is a device that provides buoyancy during water-based activities. An exemplary personal floatation device is a multipurpose device that: 1) forms a seat for use such as in a small water craft; 2) forms a raft for use in the water; and, 3) forms a plurality of floatation devices. An exemplary personal floatation system is suitable for providing floatation for a person, child, object and the like. An exemplary personal floatation system comprises a pet pad, a float, an insulated storage case, a dry storage container, and a plurality of straps. A plurality of straps may detachably attach the pet pad, the float, the insulated storage case, and the dry storage container to form the seat and the raft portions of the personal floatation device.

These together with additional objects, features and advantages of the personal floatation device will be readily apparent to those of ordinary skill in the art upon reading the following detailed description of the presently preferred, but nonetheless illustrative, embodiments when taken in conjunction with the accompanying drawings.

A personal floatation device may be configured with a pet pad attached to the horseshoe shaped float and may form a seat that can be used on the water, on land or on another watercraft, such as a wake board, a standing paddle board, boat and the like.

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In this respect, before explaining the current embodiments of the personal floatation device in detail, it is to be understood that the personal floatation device is not limited in its applications to the details of construction and arrangements of the components set forth in the following description or illustration. Those skilled in the art will appreciate that the concept of this disclosure may be readily utilized as a basis for the design of other structures, methods, and systems for carrying out the several purposes of the personal floatation device.

It is therefore important that the claims be regarded as including such equivalent construction insofar as they do not depart from the spirit and scope of the personal floatation device. It is also to be understood that the phraseology and terminology employed herein are for purposes of description and should not be regarded as limiting.

An exemplary horseshoe shaped personal floatation device can be used in a number of ways to support or float a person on the water. A person may wrap the opening of the horseshoe shaped personal floatation device around their waist or torso, and extend their arms over the left and right armrest to float in the water. A beverage may be retained in the left or right cup holder while floating in the under-arm floatation configuration. A user may use the floatation device in saddle float configuration, wherein the floatation device is configured with the closed end between their legs and under their crotch and with the one of the left and right armrests extending up in front of them and the other extending up along their back. In the saddle float orientation, a beverage may be retained in the end cup holder of the armrest extending up in front of the person. A person may use the personal floatation device in a hammock float configuration similar to the saddle-float configuration except that they lay back to extend the left armrest away from the right armrest. The armrest extending along their back opens up with respect to the closed end, wherein the length axis of the armrest is open from a parallel orientation with the other armrest to an angle of about 50 degrees or more, about 60 degrees or more, about 80 degrees more, about 100 degrees or more or about 120 degrees or more or from about 50 degrees to 140 degrees. A person may use the personal floatation device in a back-float configuration similar to the hammock float configuration except that they are in a supine orientation, with the floatation device further extended into an open configuration wherein the length axis of the left and right armrest extend at an angle of about 100 degrees or more, about 120 degrees or more, about 140 degrees more, about 160 degree or from about 100 degrees to about 180 degrees.

An exemplary personal floatation device is made from an elastomeric material, or a material that can be deformed by a load and then substantially return to an original shape after removal of the load. The exemplary personal floatation device may be deformed by opening the left and/or right armrest to enlarge the opening therebetween, and then return substantially to the original horseshoe shape. An exemplary personal floatation device, pet pad and/or dry storage case may be relatively soft, having a shore A value of about 10 to 60, with a preferred range of between about 10 to 35, or no more than about 50. An exemplary personal floatation, device, pet pad and/or dry storage case are buoyant and float on water and may be a foam, such as a closed cell foam or may include a bladder that has an internal volume of air or lightweight material, such as a foam. In an exemplary embodiment, a bladder is an inflatable bladder that can be filled through a fill port, such as a nozzle that can be filled by blowing into the nozzle. An exemplary foam is a water-

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resistant foam and may be made out of a plastic or polymeric material, such as Ethylene-Vinyl Acetate (EVA), urethane, low density polyethylene (LDPE), polyimide, polypropylene, polyvinyl chloride, silicone and the like. An exemplary personal floatation device, pet pad and/or dry storage case may comprise a cover layer that extends over the foam and is water impermeable. This cover layer may prevent water from entering into the cells of foam and reducing buoyance overtime. A cover layer may be a skin of the polymer used to make the foam and may be formed in-situ during the molding process of the floatation device. A cover may be a removable cover that can be removed and washed or interchanged for a different cover that may have a different print pattern or design, for example. A cover may be detachably attachable over the personal floatation device and may be configured to encase the personal floatation device therein, wherein at least 90 percent or more of the entire personal floatation device is contained within the cover.

An exemplary personal floatation device may have a PFD strap system comprising a PFD retaining strap that has an extending strap that is coupled to one side or armrest of the horseshoe shaped floatation device and is configured to detachably attach or couple with the opposing armrest. An extending strap may have hook-and-loop fastener that is configured to detachably attach with hook-and-loop fastener on the opposing side or armrest, for example. An exemplary PFD strap system may comprise a left PFT extending strap and a right PFD extending strap, each having a buckle to secure the horseshoe shaped float about a person. The strap can be buckled and tightened by pulling on the strap of one or both of the PFD buckles to synch the extended ends of the left armrest and right armrest tight about a person's torso. The PFD retaining strap is secured to the horseshoe shaped float by over-straps, or straps that extend over and around the horseshoe shaped float, such as around the left and/or right armrest or around the closed end, such as on either side of the neck extension. A person can be securely supported in the water with the PFD retaining strap buckled and pulled tight to secure the horseshoe shaped float about a person's torso. Note that a buckle may be on only one of the extending straps and may this buckle may be configured to couple with a buckle on the opposing armrest, such as to a buckle coupled to one of the over-straps, for example.

An exemplary personal floatation system may comprise a cover configured to extend over the personal floatation device that is detachably attachable over the personal floatation device or float. The personal floatation device may be configured to fit within the cover. The cover may be a waterproof cover or may allow for water penetration there-through. The cover may be a fabric, such as a woven fabric or a plastic cover. The cover may have a design printed thereon and a user may want to change the cover depending on the occasion, July 4th, a birthday, etc. The cover has a seam that may be configured to open for removal and insertion of the personal floatation device. The seam may be configured with a seam closure, such as a hook-and-loop fastener or with a zipper, for example. The seam may extend around the personal floatation device from the right arm bulb, down past the the right arm free end, along the right armrest or right arm extension, around the closed end and then along the left armrest, to the left arm free end, and around a portion of the left arm bulb. The seam may extend around the free ends of the left and right arm to allow easier insertion and removal of the personal floatation device. The seam may extend proximal to the bottom of the personal floatation device.

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The personal floatation device may be configured with a chin depression in the top of the neck extension. The chin depression may be a concave depression in the top of the neck extension. This chin depression may allow a person to float with their head or neck extended over or on the neck depression with their torso between the arm extensions and their arms extending back over the closed end and resting over the armrests, a reverse arm float. The depth of the depression may be about 10 mm or more, about 20 mm or more, about 30 mm or more, about 50 mm or more and any range between and including the values provided.

The personal floatation device may be configured with a second left arm groove and second right arm groove which may be configured proximal to the left arm free end and right arm free end, or just back from the left arm cup cylinder and right arm cup cylinder, respectively. These grooves may allow a strap to be configured around the two armrests to prevent them from opening up to provide more support in certain floating positions. It may also make it easier to carry and store the personal floatation device system.

The personal floatation device system may be configured with strap-loops. The strap-loops may be configured along the armrest and along the closed end of the personal floatation device and/or on a cover configured to fit over the personal floatation device, as described herein. Exemplary strap-loops may be coupled to the cover and/or directly to the personal floatation device. The left armrest strap-loop may extend along the outside of the left armrest and have a first connected end and a second connected end to form a loop that is connected with the cover or directly with the personal floatation device. The right armrest strap-loop may extend along the outside of the right armrest and have a first connected end and a second connected end to form a loop that is connected with the cover or directly with the personal floatation device. The closed end strap-loop may extend along the outside of the closed end and have a first connected end and a second connected end to form a loop that is connected with the cover or directly with the personal floatation device.

A personal floatation device may be an inflatable personal floatation device wherein a port is configured to introduce air into the inflatable personal floatation device. A desired amount of air may be introduced into the inflatable personal floatation device for comfort and personal choice. More air will make the inflatable personal floatation device harder and less air will make the inflatable personal floatation device softer or more easily deformed. An exemplary PFD extending strap, may be referred to herein as simply an extending strap.

The summary of the invention is provided as a general introduction to some of the embodiments of the invention and is not intended to be limiting. Additional example embodiments including variations and alternative configurations of the invention are provided herein.

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS

The accompanying drawings are included to provide a further understanding of the invention and are incorporated in and constitute a part of this specification, illustrate embodiments of the invention, and together with the description serve to explain the principles of the invention.

FIG. 1 is a perspective view of an embodiment of the disclosure.

FIG. 2 is a perspective view of an embodiment of the disclosure.

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FIG. 3 is a perspective view of an embodiment of the disclosure.

FIG. 4 is a side view of an embodiment of the disclosure.

FIG. 5 is a side view of an embodiment of the disclosure.

FIG. 6 is an in use view of an embodiment of the disclosure.

FIG. 7 is an in use view of an embodiment of the disclosure.

FIG. 8 is a view of an embodiment of the disclosure.

FIG. 9 is an in use view of an embodiment of the disclosure.

FIG. 10 shows an exemplary personal floatation device in use with a person floating in an underarm-floating configuration.

FIG. 11 shows an exemplary personal floatation device in use with a person floating in a saddle-floating configuration.

FIG. 12 shows an exemplary personal floatation device in use with a person floating in a hammock-floating configuration.

FIG. 13 shows an exemplary personal floatation device in use with a person floating in a back-floating configuration.

FIG. 14 shows an exemplary personal floatation device in use with a person resting in lounge-configuration.

FIG. 15 shows an exemplary personal floatation device in use with a cooler floating in a gear-float configuration.

FIG. 16 shows a perspective view of an exemplary personal floatation device having a left and right arm bulb with an end cup holder and with a cover configured over the personal floatation device.

FIG. 17 shows a front view of an exemplary personal floatation device having a left and right arm bulb with an end cup holder and with a cover configured over the personal floatation device.

FIG. 18 shows a top view of an exemplary personal floatation having a left and right arm bulb with an end cup holder and with a cover configured over the personal floatation device.

FIG. 19 shows a perspective view of an exemplary personal floatation device having a PFD strap system comprising a retaining strap that extends from the left armrest and right armrest, each having buckles to secure the PFD retaining strap together to secure the personal floatation device about a person's torso.

FIG. 20 shows a back perspective view of the exemplary personal floatation device shown in FIG. 20 and the back extension of the retaining strap extending between left and right over-straps and left and right over-straps extending around the left and right armrest in the left and right grooves, respectively.

FIG. 21 shows a perspective view of an exemplary personal floatation device being secured around a man's torso by the PFD strap system.

FIG. 22 shows a perspective view of an exemplary personal floatation device secured around a man's torso by the PFD strap system and securely keeping the man afloat in the water.

FIG. 23 shows a perspective view of an exemplary personal floatation device secured around a man's torso by the PFD strap system and securely keeping the man afloat in the water.

FIG. 24 shows a top view of an exemplary personal floatation having a left and right arm bulb with an end cup holder.

Corresponding reference characters indicate corresponding parts throughout the several views of the figures. The figures represent an illustration of some of the embodiments of the present invention and are not to be construed as

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limiting the scope of the invention in any manner. Further, the figures are not necessarily to scale, some features may be exaggerated to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present invention.

DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENTS

As used herein, the terms "comprises," "comprising," "includes," "including," "has," "having" or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Also, use of "a" or "an" are employed to describe elements and components described herein. This is done merely for convenience and to give a general sense of the scope of the invention. This description should be read to include one or at least one and the singular also includes the plural unless it is obvious that it is meant otherwise.

Certain exemplary embodiments of the present invention are described herein and are illustrated in the accompanying figures. The embodiments described are only for purposes of illustrating the present invention and should not be interpreted as limiting the scope of the invention. Other embodiments of the invention, and certain modifications, combinations and improvements of the described embodiments, will occur to those skilled in the art and all such alternate embodiments, combinations, modifications, improvements are within the scope of the present invention.

The following detailed description is merely exemplary in nature and is not intended to limit the described embodiments of the application and uses of the described embodiments. As used herein, the word "exemplary" or "illustrative" means "serving as an example, instance, or illustration." Any implementation described herein as "exemplary" or "illustrative" is not necessarily to be construed as preferred or advantageous over other implementations. All of the implementations described below are exemplary implementations provided to enable persons skilled in the art to practice the disclosure and are not intended to limit the scope of the appended claims.

A personal floatation system includes a personal floatation device (PFD) that is adapted for use with a person to provide buoyance for a person in the water and may keep a person afloat in the water. The personal floatation system may be adapted for use with a pet. The PFD is configured for use with water. The PFD is a device that provides buoyancy during water-based activities. The personal floatation system may be a multipurpose system that: 1) forms a seat for use in a small water craft; 2) forms a raft for use in the water; and, 3) forms a plurality of floatation devices. The personal floatation system may include a pet pad, a float, or horseshoe shaped float, an insulated storage case, a dry storage case, and a plurality of straps. The plurality of straps may be configured to interconnect the pet pad, the float, the insulated storage case, and the dry storage case to form the seat and the raft portions. A person may use the float as a PFD in an emergency situation. A pet may use the pet pad as a PFD in an emergency situation.

The person 201 refers to an individual who is using the PFD as intended. The pet 202 refers to a domesticated animal that is accompanying the person 201 in the small

water **204** craft **203** and the water **204**. The small water **204** craft **203** refers to a small vehicle adapted for use in travel over water **204**. The water **204** refers to the body of water **204** on which the personal floatation system is being used. The beverage **205** refers to a can, cup or bottle of a consumable liquid.

The exemplary pet pad **101** is a rectangular plate, or panel, molded from a foam, such as a closed cell ethylene-vinyl acetate (EVA) foam. The EVA foam used in the pet pad **101** provides the buoyancy of the pet pad **101**. When the personal floatation system **100** is used as a raft, as shown most clearly in FIG. 9, the pet pad **101** forms a buoyant horizontal surface that forms a seat that keeps the person **201** afloat in the water **204**. When the personal floatation system **100** is used as a PFD as shown most clearly in FIG. 7, the pet pad **101** forms a buoyant horizontal surface that keeps the pet **202** afloat in the water **204**. When the personal floatation system **100** is used as a seat as shown most clearly in FIGS. 1 and 6, the pet pad **101** forms a pedestal that supports the float **102** and the insulated storage case **103** on the small water **204** craft **203**.

The exemplary pet pad **101** comprises a buoyant plate **110**. The buoyant plate **110** further comprises a first slot **111**, a second slot **112**, and a third slot **113**.

The exemplary buoyant plate **110** is a rectangular plate formed from EVA foam. The buoyant plate **110** provides the buoyancy of the pet pad **101**. The first slot **111** is an aperture formed through the buoyant plate **110**. The first slot **111** is sized such that a strap selected from the plurality of straps **105** can be inserted through the first slot **111**. The second slot **112** is an aperture formed through the buoyant plate **110**. The second slot **112** is sized such that a strap selected from the plurality of straps **105** can be inserted through the second slot **112**. The third slot **113** is an aperture formed through the buoyant plate **110**. The third slot **113** is sized such that a strap selected from the plurality of straps **105** can be inserted through the third slot **113**.

The exemplary float **102** and the insulated storage case **103** are mounted on the pet pad **101** when the personal floatation system **100** is used as the seat. The float **102** is a horseshoe shaped structure formed from EVA foam. The EVA foam used in the float **102** provides the buoyancy of the float **102**.

When the exemplary personal floatation system **100** is used as a raft, as shown most clearly in FIG. 9, the float **102** forms a buoyant structure **121** that further keeps the person **201** afloat in the water **204**. In this scenario, the float **102** further forms a left armrest **161** and a right armrest **171** for use by the person **201**. When the personal floatation system **100** is used as a PFD as shown most clearly in FIG. 7, the float **102** forms a buoyant structure **121** that is worn by the person **201** and that keeps the person **201** afloat in the water **204**. When the personal floatation system **100** is used as a seat as shown most clearly in FIGS. 1 and 6, the float **102** forms a backrest, a left armrest **161**, and a right armrest **171** for use by the person **201**.

The exemplary personal floatation device **120**, or float **102** comprises a buoyant structure **121** and a bias mass **122**. The buoyant structure **121** is further defined with a flush or planar bottom surface **224**, a structured surface **225**, upper or top surface, a negative space **221**, and an inner perimeter **222**. The negative space **221** refers to the area partially bounded by the inner perimeter **222** of the float **102**. The inner perimeter **222** refers to the concave surfaces of the float **102** formed by the left armrest **161**, the right armrest **171**, and the right armrest **171**. The flush surface **224** is the surface of the float **102** that is proximal to the pet pad **101**

when the personal floatation system **100** is used as a raft and a seat. The flush surface **224** forms a planar surface such that the flush surface **224** sits flush against the pet pad **101**. The structured surface **225** is the surface of the float **102** that is distal from the flush surface **224**.

The exemplary buoyant structure **121** is a horseshoe shaped structure formed from a buoyant material, such as an EVA foam. The buoyant structure **121** forms the characteristic horseshoe shape of the float **102**. The buoyant structure **121** comprises a left armrest **161**, a right armrest **171**, and a closed end **181**. The left armrest **161** is further defined with a left arm free end **164**. The right armrest **171** is further defined with a right arm free end **174**. The closed end **181** is further defined with a closed end midpoint **182**.

The exemplary left armrest **161** comprises a left arm groove **162** and a left arm cup holder **163**. The left arm cup holder **163** further comprises a left arm bulb **165** and a left arm cup cylinder **166**. The right armrest **171** comprises a right arm groove **172** and a right arm cup holder **173**. The right arm cup holder **173** further comprises a right arm bulb **175** and a right arm cup cylinder **176**.

The exemplary left armrest **161** is the arm of the horseshoe shaped float **106** of the float **102** that is proximal to the left side of the person **201**, or extends from the right side of the closed end when viewed from the extended ends of the left and right armrests. The left arm groove **162** is a groove formed in the structured surface **225** of the left armrest **161**. The left groove **162** receives the first hook-and-loop fastener **151** that the first hook-and-loop fastener **151** will not shift use. The left arm cup holder **163** is a structure that is arm such during formed at the left arm free end **164**. The left arm cup holder **163** provides a location to store a beverage **205** during water **204** based activities. The left arm free end **164** is the end of the left armrest **161** that is distal from the closed end **181**.

The exemplary left arm bulb **165** is a roughly cylindrical structure that attaches to the left arm free end **164** of the left armrest **161**. The left arm cup cylinder **166** is a negative space that is formed as a cylinder. The left arm cup cylinder **166** is coaxially located within the left arm bulb **165**. The opening to the left arm cup cylinder **166** is formed in the structured surface **225** of the left armrest **161**. The left arm cup cylinder **166** is sized to receive the beverage **205**.

The exemplary right armrest **171** is the arm of the horseshoe shape of the float **102** that is proximal to the right side of the person **201**. The right arm groove **172** is a groove formed in the structured surface **225** of the right armrest **171**. The right arm groove **172** receives the third hook-and-loop fastener **153** such that the third hook-and-loop fastener **153** will not shift during use. The right arm cup holder **173** is a structure that is formed at the right arm free end **174**. The right arm cup holder **173** provides a location to store a beverage **205** during water **204** based activities. The right arm free end **174** is the end of the right armrest **171** that is distal from the closed end **181**.

The exemplary right arm bulb **175** is a roughly cylindrical structure that attaches to the right arm free end **174** of the right armrest **171**. The right arm cup cylinder **176** is a negative space that is formed as a cylinder. The right arm cup cylinder **176** is coaxially located within the right arm bulb **175**. The opening to the right arm cup cylinder **176** is formed in the structured surface **225** of the right armrest **171**. The right arm cup cylinder **176** is sized to receive the beverage **205**.

The exemplary bias mass **122** is an additional structure formed from EVA foam. The bias mass **122** attaches to the structured surface **225** of the buoyant structure **121** at the

closed end midpoint **182** of the closed end **181**. The bias mass **122** creates an asymmetry in the buoyancy of the float **102**. When the personal floatation system **100** is worn as a PFD the asymmetry formed by the bias mass **122** ensures that the head of the person **201** remains above the water **204**.

The exemplary insulated storage case **103** is an insulating structure suitable for storing foodstuffs and beverages **205**. The insulated storage case **103** is a hollow structure formed in a roughly rounded rectilinear block shape. The insulated storage case **103** is formed from EVA foam. The EVA foam used in the insulated storage case **103** provides the insulated storage case **103** with enough buoyancy that the insulated storage case **103** can keep itself and the contents contained within the insulated storage case **103** afloat. The form factor of the insulated storage case **103** is congruent to the form factor of the hollow interior of the float **102** such that the insulated storage case **103** will fit flush within the hollow center of the float **102**. In this configuration the insulated storage case **103** and the float **102** will effectively form a single structure.

When the personal floatation system **100** is used as a raft, as shown most clearly in FIG. 9, the insulated storage case **103** floats beside the person **201** in the water **204**. When the personal floatation system **100** is used as a PFD as shown most clearly in FIG. 7, the insulated storage case **103** floats beside the person **201** in the water **204**. When the personal floatation system **100** is used as a seat as shown most clearly in FIGS. 1 and 6, the insulated storage case **103** forms the bench upon which the person **201** sits while in the small water **204** craft **203**.

The exemplary insulated storage case **103** comprises a container **131**, a lid **132**, a hinge **133**, a handle slot **134**, and a plurality of bungee cords **135**. The container **131** is further defined with an outer perimeter **223**. The outer perimeter **223** refers to the exterior surfaces of the container **131** of the insulated storage case **103** that are congruent to the inner perimeter **222** of the float **102**.

The exemplary container **131** is a hollow insulating structure formed from EVA foam. The form factor of the container **131** is congruent to the form factor of the hollow interior of the buoyant structure **121**. The container **131** receives the foodstuffs, the beverages **205**, and other contents stored in the container **131** through the open face. The EVA foam used in the container **131** ensures that the insulated storage case **103** and the contents of the insulated storage case **103** remain afloat while in the water **204**.

The exemplary lid **132** is a cover that is attached to the container **131** such that the lid **132** rotates relative to the open face of the container **131**. The lid **132** is a barrier that covers the open face of the container **131**. The lid **132** controls access into the hollow interior of the container **131**.

The exemplary hinge **133** refers to commercially available hardware that attaches the lid **132** to the container **131** such that the lid **132** rotates in a manner that opens and closes the open face of the container **131**. The handle slot **134** is an aperture that forms a channel in a wall of the container **131**. The handle slot **134**: 1) forms a grip used by the person **201**; and 2) forms an anchor point that allows the second hook-and-loop fastener **152** to attach the insulated storage case **103** to the float **102**. Each of the plurality of bungee cords **135** attaches to the exterior surface of the container **131**. Each of the plurality of bungee cords **135** is an elastic cord used to attach objects to the exterior surface of the container **131**.

The exemplary dry storage case **104** is a water **204** tight containment structure. The dry storage case **104** contains items that the person **201** wants to keep dry while on the

water **204**. When the personal floatation system **100** is used as a raft, as shown most clearly in FIG. 9, the dry storage case **104** attaches to the float **102**. When the personal floatation system **100** is used as a PFD as shown most clearly in FIG. 7, the dry storage case **104** attaches to the float **102**. When the personal floatation system **100** is used as a seat as shown most clearly in FIGS. 1 and 6, the dry storage case **104** attaches to the float **102**.

The exemplary dry storage case **104** comprises a dry bag **141** and a fourth hook-and-loop fastener **154**. The fourth hook-and-loop fastener **154** further comprises a seventh hook/loop surface **197** and an eighth hook/loop surface **198**. The dry bag **141** is a commercially available bag that is designed to contain objects in a water **204** impermeable environment. The dry bag **141** attaches to the float **102** using the fourth hook-and-loop fastener **154**. The seventh hook/loop surface **197** of the fourth hook-and-loop fastener **154** attaches to the back of the closed end **181** of the float **102** at the closed end midpoint **182**. The eighth hook/loop surface **198** attaches to the dry bag **141**. The dry bag **141** attaches to the float **102** by pressing the seventh hook/loop surface **197** to the eighth hook/loop surface **198** to form the fourth hook-and-loop fastener **154**.

Each of the plurality of straps **105** is a commercially available fastening device. When the personal floatation system **100** is used as a raft, as shown most clearly in FIG. 9, the plurality of straps **105** binds the float **102** to the pet pad **101**. When the personal floatation system **100** is used as a PFD as shown most clearly in FIG. 7, the plurality of straps **105** binds the float **102** to the person **201**. When the personal floatation system **100** is used as a seat as shown most clearly in FIGS. 1 and 6, the plurality of straps **105** binds the float **102** and the insulated storage case **103** to the pet pad **101**.

The plurality of straps **105** comprises a first hook-and-loop fastener **151**, a second hook-and-loop fastener **152**, and a third hook-and-loop fastener **153**. The first hook-and-loop fastener **151** further comprises a first hook/loop surface **191**, a second hook/loop surface **192**, and a first seam **211**. The second hook-and-loop fastener **152** further comprises a third hook/loop surface **193**, a fourth hook/loop surface **194**, and a second seam **212**. The third hook-and-loop fastener **153** further comprises a fifth hook/loop surface **195**, a sixth hook/loop surface **196**, and a third seam **213**.

The first hook-and-loop fastener **151** is a strap configured as a hook-and-loop fastener. The first hook-and-loop fastener **151** is formed by attaching the first hook/loop surface **191** to the second hook/loop surface **192** using the first seam **211**. The first hook/loop surface **191** and the second hook/loop surface **192** are joined back-to-back such that the first hook-and-loop fastener **151** will form a loop when the first hook/loop surface **191** is pressed against the second hook/loop surface **192**.

The second hook-and-loop fastener **152** is a strap configured as a hook-and-loop fastener. The second hook-and-loop fastener **152** is formed by attaching the third hook/loop surface **193** to the fourth hook/loop surface **194** using the second seam **212**. The third hook/loop surface **193** and the fourth hook/loop surface **194** are joined back-to-back such that the second hook-and-loop fastener **152** will form a loop when the third hook/loop surface **193** is pressed against the fourth hook/loop surface **194**.

The third hook-and-loop fastener **153** is a strap configured as a hook-and-loop fastener. The third hook-and-loop fastener **153** is formed by attaching the fifth hook/loop surface **195** to the sixth hook/loop surface **196** using the third seam **213**. The fifth hook/loop surface **195** and the sixth hook/loop surface **196** are joined back-to-back such that the third

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hook-and-loop fastener **153** will form a loop when the fifth hook/loop surface **195** is pressed against the sixth hook/loop surface **196**.

The first seam **211** is a sewn seam. The second seam **212** is a sewn seam. The third seam **213** is a sewn seam. Sewn seams are discussed in greater detail elsewhere in this disclosure.

The first hook-and-loop fastener **151** attaches the left armrest **161** of the float **102** to the pet pad **101** during use of the personal floatation system **100** as a raft or a seat. The first hook-and-loop fastener **151** is placed over the left arm groove **162** of the left armrest **161**, threaded through the first slot **111** and looped around the edge of the buoyant plate **110**. The first hook-and-loop fastener **151** is secured by pressing the first hook/loop surface **191** against the second hook/loop surface **192**. The first hook-and-loop fastener **151** binds the left armrest **161** to the left arm of the person **201** when the personal floatation system **100** is used as a PFD. The first hook-and-loop fastener **151** is secured by pressing the first hook/loop surface **191** against the second hook/loop surface **192**.

The third hook-and-loop fastener **153** attaches the right armrest **171** of the float **102** to the pet pad **101** during use of the personal floatation system **100** as a raft or a seat. The third hook-and-loop fastener **153** is placed over the right arm groove **172** of the right armrest **171**, threaded through the third slot **113** and looped around the edge of the buoyant plate **110**. The third hook-and-loop fastener **153** is secured by pressing the fifth hook/loop surface **195** against the sixth hook/loop surface **196**. The third hook-and-loop fastener **153** binds the right armrest **171** to the right arm of the person **201** when the personal floatation system **100** is used as a PFD. The third hook-and-loop fastener **153** is secured by pressing the fifth hook/loop surface **195** against the sixth hook/loop surface **196**.

The second hook-and-loop fastener **152** attaches the closed end **181** of the float **102** to the pet pad **101** during use of the personal floatation system **100** as a raft or a seat. The second hook-and-loop fastener **152** further attaches the container **131** of the insulated storage case **103** to the pet pad **101** during use of the personal floatation system **100** as a raft or a seat. The second hook-and-loop fastener **152** is placed over the closed end **181** of the float **102**. The second hook-and-loop fastener **152** is threaded through the handle slot **134** of the insulated storage case **103**. The second hook-and-loop fastener **152** is threaded through the second slot **112** and looped around the edge of the buoyant plate **110**. The second hook-and-loop fastener **152** is secured by pressing the third hook/loop surface **193** against the fourth hook/loop surface **194**.

As shown in FIG. 9, a person **201** is floating on the personal floatation device **120** and is sitting on the pet pad **101** that is retained to the buoyant structure **121**, the horseshoe shaped float, by straps **105**. An insulated storage case **103** may be tethered to the personal floatation device and carry beverages.

As shown in FIG. 10, an exemplary personal floatation device **120** is in use with a person **201** floating in an underarm-floating configuration. The person is floating in the horseshoe shaped floatation device and has the left armrest **161** and right armrest **171** configured under their left and right arms, respectively, and their torso is configured within the opening or space between the two armrests. The person's back is resting against the closed end **181** and their head can lean back against the neck extension **184**. A beverage **205** is configured in the right cup cylinder **176**. Note that the person could turn around with their chest

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against the closed end and place their chin on the neck extension **184**. The right armrest **171** extends to right armrest free end having a cylindrically shaped right arm bulb **175** configured with a right arm cup cylinder **176** and a right arm end cup holder **178**. The left armrest **161** extends to left armrest free end having a cylindrically shaped left arm bulb **165** configured with a left arm cup cylinder **166** and a left arm end cup holder **168**.

As shown in FIG. 11, an exemplary personal floatation device **120** is in use with a person **201** floating in a saddle-floating configuration. The closed end **181** is configured between the person's legs and extends through their crotch. The left armrest **161** extends from the crotch up along the person's back and the right armrest **171** extends from the crotch in front of the person. A beverage **205** is configured in the right end cup holder **178** of the right arm bulb **175**.

As shown in FIG. 12, an exemplary personal floatation device **120** is in use with a person **201** floating in a hammock-floating configuration. The hammock-floating configuration is similar to the saddle-floating configuration except that the person is more reclined, and the personal floatation device is extended or opened up to expand the distance between the left and right arm bulbs. The closed end **181** is more elongated or opened up and the inclusive angle **179** between the right arm length axis **170** and the left arm length axis **160** is about 60 degrees or more or about 80 degrees or more, about 90 degrees or more, about 110 degrees or more and any range between and including the inclusive angles provided. Put another way, the right armrest length axis **170** extends at an offset from vertical, such as at least 20 degrees from vertical, to as much as 60 degrees from vertical or 30 degrees from horizontal. The personal floatation device is configured between the person's legs and extends through the crotch. The right armrest **171** extends from the crotch up along the person's back and the left armrest **161** extends from the crotch in front of the person. A beverage **205** is configured in the left end cup holder **168** of the left arm bulb **165**.

As shown in FIG. 13, an exemplary personal floatation device **120** is in use with a person **201** floating in a back-floating configuration. The back-floating configuration is similar to the hammock floating configuration except that the person is in a supine and fully reclined position and the personal floatation device is extended to a substantially linear configuration. The closed end **181** is substantially straightened or elongated to a substantially straight orientation, or within about 20 degrees of straight. The inclusive angle **179** between the right arm length axis **170** and the left arm length axis **160** is about 120 degrees or more or about 140 degrees or more, about 160 degrees or more, about 180 degrees or more and any range between and including the inclusive angles provided. Put another way, the right armrest length axis **170** extends at an offset from vertical, such as at least 60 degrees from vertical, to as much as 90 degrees from vertical or substantially horizontal. The personal floatation device is configured between the person's legs and extends through the crotch. The right armrest **171** extends along the person's back and the left armrest **161** extends from the crotch in front of the person. A beverage **205** is configured in the end cup holder **168** of the left arm bulb **165**.

As shown in FIG. 14, an exemplary personal floatation device **120** in use with a person **201** resting in lounge-configuration. The person is resting with their back in the open area between the left and right armrests and has their neck and/or head on the closed end. The person's arms are

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resting comfortably on the left and right armrests. A beverage **205** is retained in the left cup holder **166**.

FIG. **15** shows an exemplary personal floatation device in use with a insulated storage case **103**, or cooler **206**, floating in a gear-float configuration. The cooler **206** is retained in the open area between the left armrest **161** and right armrest **171** and is secured on the pet pad **101**. The pet pad is retained to the horseshoe shaped float **106** by the straps **151** and **153**. As shown in FIG. **3**, the straps extend down through the slots in the pet pad **101**.

Referring now to FIGS. **16** to **18**, an exemplary personal floatation device **120** has a left armrest **161** and right armrest **171** that extend from the closed end to produce a horseshoe shaped floatation device having an opening between the two extended ends of the left and right armrests, or between the two bulbs configured on the extended ends of the left and right armrest. The left and right armrests have grooves **162**, **172**, respectively to retain a strap. Each of the left and right arm bulbs **165**, **175**, respectively have a cup cylinder **166**, **176** for receiving and retaining a beverage vertically, when the personal floatation device is used horizontally, as shown in FIGS. **10** and **14**, and an end cup holder **168**, **178**, for receiving and retaining a beverage when the personal floatation device is used in an extended configuration, as shown in FIGS. **11** to **13**. The exemplary personal floatation device also has a neck extension **184**, that extends up vertically from the closed end, to provide support for a user's head. The personal floatation device has a top **124** and bottom **126**. The bottom **126** is planar, or flat to allow the personal floatation device to rest securely on a flat surface, such as a paddle board, as shown in FIG. **14**, or on the pet pad **101**, as shown in FIG. **1**.

As shown in FIGS. **16-18**, **20** and **21**, a cover **410** is shown configured over the personal floatation device **120** and the seam **400** of the cover extends from the left arm bulb **165**, along the left armrest **161**, around the closed end **181**, back down along the right armrest **171** and around a portion of the right arm bulb **175**. The seam **400** of the cover **410** extends around the bottom perimeter of the personal floatation device, as shown to prevent contact with the seam when floating within the personal floatation device.

As shown in FIG. **17**, the horseshoe shaped personal floatation device is made of a foam **186** having a cover layer **188** that extends substantially over the entire outer surface, or at least 90% or more and preferably 95% or more of the outer surface area, and prevents water from entering into the cells of the foam, or has no bulk water flow therethrough, a liquid impermeable layer. The end cup holders **168** and **178** have an opening from the extended end of the bulb **165**, **175**, respectively, to the left and right cup cylinders **166**, **176**, respectively. The opening extends down a depth from the top of the personal floatation device but may also extend from the bottom surface or be an aperture having between the top and bottom surfaces. As shown in FIG. **17**, the cup holders **163**, **173** are dual cup holders, or cup holders that can retain a beverage in one of two different orientations that are substantially orthogonal to each other. A beverage can be retained in a vertical orientation or extending vertically in the cup cylinder or orthogonal to vertical, or horizontal or in plane with the top surface and within the end cup holder.

As shown in FIG. **18**, the left and right bulbs each have a cup holder **163**, **173** that are dual cup holders, or are configured to retain a beverage in orthogonal directions. The right cup cylinder **176** has a diameter **177**, and the end cup holder **178** is an opening to the cup cylinder that extends an arc or radius **179**. The diameter and radius may be about the same dimension and configured to hold a standard size

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beverage can, such as about 50 mm or more, about 75 mm or more, about 100 mm or more, about 75 to 125 mm and any range between and including the values provided. The opening may be smaller than an average beverage container and the resilient nature may enable the opening to expand to receive the beverage container. The left armrest length axis **160** and right armrest length axis **170** extend substantially parallel, or within about 20 degrees or parallel with each other, as shown in FIG. **18**. When the left and/or right armrest are opened to expand the opening **180**, between the left and right armrest, the length axes will extend at an angle from an intersection point with each other. This angle may be about 120 degrees when used in a hammock-float configuration and closer to 180 degrees when used in a back-float configuration.

Referring now to FIGS. **19** to **23**, an exemplary personal floatation device **120** has a PFD strap system **300** comprising a PFD retaining strap **310** that has a left PFT extending strap **322** and a right PFD extending strap **332** having a buckles **325**, **335** respectively, to secure the horseshoe shaped float **106** about a person. The strap can be buckled and tightened by pulling on the strap extending from the left PFD buckle **325** to synch the extended ends of the left armrest **161** and right armrest **171** tight about a person. The PFD retaining strap is secured to the horseshoe shaped float **106** by left groove over-strap **382** and right groove over-strap **392**, that extend over and around the left groove **162** and right groove **172**, of the left armrest **161** and right armrest **171**, respectively. The PFD retaining strap is also secured to the horseshoe shaped float by left over-strap **362** and right over-strap **372** that extend over and around a portion of the closed end **181**, or on opposing sides and proximal to the neck extension **184**. Note that a single extending strap, such as the left extending strap **322** may extend over to the right arm and be detachably attached to a buckle coupled to the PFD strap system or to the PFD itself. For example, the buckle of the left extending strap may buckle to a buckle coupled to the right groove over-strap **392**. Alternatively, the strap may detachably attach via hook-and-loop fastener material. The left extending strap may have hook-and-loop fastener proximal the end of the strap and it may be pulled tight and coupled with hook-and-loop fastener on the horseshoe shaped float **106**, or to the PFD strap system, such as to the right extending strap.

As shown in FIG. **20**, the left over-strap **362** and right over-strap **372** have a left over-strap buckle **365** and right over-strap buckle **375**, respectively. Likewise, the left groove over-strap **382** and right groove over-strap **392** may have buckles to enable the PFD strap system **300** to be removed from the horseshoe shaped float **106**. A back extension strap **318**, which is part of the PFD retaining strap **310**, is shown in FIG. **20**. The cover **410** is shown configured over the personal floatation device **120** and the seam **400** of the cover extends along the closed end **181**. The seam **400** of the cover **410** extends around the bottom perimeter of the personal floatation device around the closed end.

As shown in FIG. **21**, the exemplary personal floatation device **120** has the PFD strap system **300** buckled by the left PFD buckle **325** and right PFD buckle **335**. The left arm bulb **165** and right arm bulb **175** are pulled together by the strap to secure the personal floatation device about the person's torso.

As shown in FIG. **22**, the person is being securely supported in the water by the personal floatation device **120** having the PFD retaining strap **310** secured about the horseshoe shaped float **106**.

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As shown in FIG. 23, the person is being securely supported in the water by the personal floatation device 120 and is facing the closed end 181 or the horseshoe shaped float 106.

Referring now to FIGS. 16 to 23, an exemplary personal floatation system 100, comprises a personal floatation device 120 having a cover 410 that is detachably attachable over the personal floatation device or float 102, wherein the personal floatation device is configured to fit within the cover. The cover may be a waterproof cover or may allow for water penetration therethrough. The cover may be a fabric, such as a woven fabric or a plastic cover. The cover may have a design printed thereon and a user may want to change the cover depending on the occasion, July 4th, a birthday, etc. The cover has a seam 400 that may be configured to open for removal and insertion of the personal floatation device 120. The seam may be configured with a seam closure 402, such as a hook-and-loop fastener or zipper chain that is closed by a zipper slider 404, for example. The seam 400 may extend around the personal floatation device 120 from the right arm free end 174, along the right armrest 171 or right arm extension, around the closed end 181 and then along the left armrest 181, to the left arm free end 184. The seam may extend around the free ends of the left and right arm to allow easier insertion and removal of the personal floatation device. The seam may extend proximal to the bottom 126 of the personal floatation device.

The personal floatation device 120 may be configured with a chin depression 185 in the top of the neck extension 184. The chin depression 185 may be a concave depression in the top of the neck extension 184, as best shown in FIG. 18. This chin depression may allow a person to float with their head and/or neck extended over or on the neck depression with their torso between the arm extensions and their arms extending back over the closed end and resting over the armrests, a reverse arm float. The depth of the depression may be about 10 mm or more, about 20 mm or more, about 30 mm or more, about 50 mm or more and any range between and including the values provided.

The personal floatation device 120 may be configured with a second left arm groove 162' and second right arm groove 172', as best shown in FIG. 16, which may be configured proximal to the left arm free end 164 and right arm free end 174, or just back from the left arm cup cylinder 166 and right arm cup cylinder 176, respectively. These grooves may allow a strap to be configured around the two armrests to prevent them from opening up to provide more support in certain floating positions. It may also make it easier to carry and store the personal floatation device system 100.

The personal floatation device system 100 may be configured with strap-loops. The strap-loops may be configured along the armrest and along the closed end of the personal floatation device 120 and/or on a cover 410. As shown in FIG. 16, the strap-loops are coupled to the cover 410. The left armrest strap-loop 460 is shown in FIG. 16 extending along the outside of the left armrest and having a first connected end 462 and a second connected end 464 to form a loop that is connected with the cover 410 configured over the personal floatation device 120. The right armrest strap-loop 470 is shown in FIG. 18 extending along the outside of the right armrest and having a first connected end 472 and a second connected end 474 to form a loop that is connected with the cover 410 configured over the personal floatation device 120. The closed end strap-loop 480 is also shown in FIG. 18 and extends along the outside of the closed end and has a first connected end 482 and a second connected end

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484 to form a loop that is connected with the cover 410 over the personal floatation device 120.

As shown in FIGS. 19 and 24, the personal floatation device 120 does not have the cover and the strap-loops are coupled or attached directly to the personal floatation device, such as the left armrest strap-loop 660 as shown in FIG. 19. As shown in FIG. 24, the left armrest strap-loop 660 is shown extending along the outside of the left armrest and having a first connected end 662 and a second connected end 664 to form a loop that is connected with the personal floatation device 120. The right armrest strap-loop 670 is shown extending along the outside of the right armrest and having a first connected end 672 and a second connected end 674 to form a loop that is connected with the cover or personal floatation device. The closed end strap-loop 680 is also shown and extends along the outside of the closed end and has a first connected end 682 and a second connected end 684 to form a loop that is connected with the personal floatation device.

The following definitions are used in this disclosure:

Anchor: As used in this disclosure, anchor means to hold an object firmly or securely.

Anchor Point: As used in this disclosure, an anchor point is a location to which a first object can be securely attached to a second object.

Bind: As used in this disclosure, to bind is a verb that means to tie or secure a first object to a second object by wrapping a third object around the first object and the second object.

Bungee: As used in this disclosure, the term bungee refers to an elastic cord or a mesh of elastic cords.

Channel: As used in this disclosure, a channel is a tubular passage through which an object or fluid is passed through.

Coaxial: As used in this disclosure, coaxial is a term that refers to a first object that is inserted or contained within a second object such: 1) that the first object and the second object share the same center point if the first object and the second object are treated as two dimensional objects; or, 2) that the first object and the second object share the same center axis if the first object and the second object are treated as a cylinder.

Congruent: As used in this disclosure, congruent is a term that compares a first object to a second object. Specifically, two objects are said to be congruent when: 1) they are geometrically similar; and, 2) the first object can be superimposed over the second object such that the first object aligns, within manufacturing tolerances, with second object.

Copolymer: As used in this disclosure, a copolymer is a polymer formed from two or more repeating molecules, also referred to as monomers.

Cord: As used in this disclosure, a cord is a long, thin, and flexible piece of string, line, rope, or wire. Cords are made from yarns, piles, or strands of material that are braided or twisted together or from a monofilament, such as fishing line. Cords have tensile strength but are too flexible to provide compressive strength and are not suitable for use in pushing objects. String, line, cable, and rope are synonyms for cord.

Correspond: As used in this disclosure, the term correspond means that a first object is in some manner linked to a second object in a one-to-one relationship.

Cylinder: As used in this disclosure, a cylinder is a geometric structure defined by two identical flat and parallel ends, also commonly referred to as bases, which are circular in shape and connected with a single curved surface, referred to in this disclosure as the face. The cross section of the cylinder remains the same from one end to another. The axis

of the cylinder is formed by the straight line that connects the center of each of the two identical flat and parallel ends of the cylinder. Unless otherwise stated within this disclosure, the term cylinder specifically means a right cylinder which is defined as a cylinder wherein the curved surface perpendicularly intersects with the two identical flat and parallel ends.

Elastic: As used in this disclosure, an elastic is a material or object that deforms when a force is applied to it and that is able to return to its relaxed shape after the force is removed. A material that exhibits these qualities is also referred to as an elastomeric material.

Elastic Cord: As used in this disclosure, an elastic cord is a cord that contains elastic yarns as some of the yarns that make up the cord. An elastic cord is constructed such that the elastic cord will stretch when a force is applied and will return to its original shape when after the force is removed.

Shock cord and bungee cord are synonyms for an elastic cord.

Ethylene-Vinyl Acetate: As used in this disclosure ethylene-vinyl acetate (CAS 24938-78-8—also referred to as polyethylene-vinyl acetate) is a copolymer formed from ethylene and vinyl acetate molecules.

Fastener: As used in this disclosure, a fastener is a device that is used to join or affix two objects. Fasteners generally comprise a first element which is attached to the first object and a second element which is attached to the second object such that the first element and the second element join to affix the first object and the second object. Common fasteners include, but are not limited to, hooks, zippers, snaps, buttons, buckles, quick release buckles, or hook-and-loop fasteners.

Flush: As used in this disclosure, the term flush is used to describe the alignment of a first surface and a second surface on a single plane.

Form Factor: As used in this disclosure, the term form factor refers to the size and shape of an object.

Geometrically Similar: As used in this disclosure, geometrically similar is a term that compares a first object to a second object wherein: 1) the sides of the first object have a one-to-one correspondence to the sides of the second object; 2) wherein the ratio of the length of each pair of corresponding sides are equal; 3) the angles formed by the first object have a one-to-one correspondence to the angles of the second object; and, 4) wherein the corresponding angles are equal.

Grip: As used in this disclosure, a grip is an accommodation formed on or within an object that allows the object to be grasped or manipulated by a hand.

Hinge: As used in this disclosure, a hinge is a device that permits the turning, rotating, or pivoting of a first object relative to a second object.

Hook-and-loop Fastener: As used in this disclosure, a hook-and-loop fastener is a fastener that comprises a hook surface and a loop surface. The hook surface comprises a plurality of minute hooks. The loop surface comprises a surface of uncut pile that acts like a plurality of loops. When the hook surface is applied to the loop surface, the plurality of minute hooks fastens to the plurality of loops securely fastening the hook surface to the loop surface. A note on usage: when fastening two objects the hook surface of a hook-and-loop fastener will be placed on the first object and the matching loop surface of a hook-and-loop fastener will be placed on the second object without significant regard to which object of the two objects is the first object and which of the two objects is the second object. When the hook surface of a hook-and-loop fastener or the loop surface of a

hook-and-loop fastener is attached to an object this will simply be referred to as the “hook/loop surface” with the understanding that when the two objects are fastened together one of the two objects will have a hook surface and the remaining object will have the loop surface.

Horseshoe: As used in this disclosure, a horseshoe refers to an object that has the shape of a horseshoe. The horseshoe refers to a metal structure that is used to protect the hoof of a horse. Specifically, the metal structure is a metal band that is attached around the perimeter of the bottom of the hoof of the horse. The horseshoe has an appearance of open U-shaped ring having a curved closed end and two extension that extend out from the closed end in the same direction.

Insulating Structure: As used in this disclosure, an insulating structure is a structure that inhibits, and ideally prevents, the transfer of heat through the insulating structure. Insulating structures may also be used to inhibit or prevent the transfer of sound through the insulating structure. Methods to form insulating structures include, but are not limited to: 1) the use of materials with low thermal conductivity; and, 2) the use of a structural design that places a vacuum within the insulating structure within the anticipated transfer path of the heat or sound.

Lid: As used in this disclosure, a lid is a removable cover that is placed over an opening of a hollow structure to enclose the hollow structure.

Loop: As used in this disclosure, a loop is a length of a first linear structure including, but not limited to, shafts, lines, cords, or ribbons, that is: 1) folded over and joined at the ends forming an enclosed space; or, 2) curved to form a closed or nearly closed space within the first linear structure. In both cases, the space formed within the first linear structure is such that a second linear structure such as a line, cord or a hook can be inserted through the space formed within the first linear structure. Within this disclosure, the first linear structure is said to be looped around the second linear structure.

Negative Space: As used in this disclosure, negative space is a method of defining an object through the use of open or empty space as the definition of the object itself, or, through the use of open or empty space to describe the boundaries of an object.

One to One: When used in this disclosure, a one-to-one relationship means that a first element selected from a first set is in some manner connected to only one element of a second set. A one-to-one correspondence means that the one-to-one relationship exists both from the first set to the second set and from the second set to the first set. A one-to-one fashion means that the one-to-one relationship exists in only one direction.

PFD: As used in this disclosure, PFD is an acronym for a personal floatation device. A personal floatation device is safety equipment in the form of a garment or device that assists a person in remaining afloat in water.

Perimeter: As used in this disclosure, a perimeter is one or more curved or straight lines that bounds an enclosed area on a plane or surface. The perimeter of a circle is commonly referred to as a circumference.

Pivot: As used in this disclosure, a pivot is a rod or shaft around which an object rotates or swings.

Plate: As used in this disclosure, a plate is a smooth, flat and semi-rigid or rigid structure that has at least one dimension that: 1) is of uniform thickness; and 2) that appears thin relative to the other dimensions of the object. Plates often have a rectangular or disk like appearance.

Rectilinear Block: As used in this disclosure, a rectilinear block refers to a three-dimensional structure comprising a

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plurality of rectangular surfaces. Rectilinear blocks are similar to rectangular blocks and are often used to create a structure with a reduced interior volume relative to a rectangular block. Within this disclosure, a rectilinear block may further comprise rounded edges and corners.

Rounded: As used in this disclosure, the term rounded refers to the replacement of an apex, vertex, or edge or brink of a structure with a generally smooth curvature wherein the concave portion of the curvature faces the interior or center of the structure.

Cylindrical shaped as used to describe the right or left arm bulb includes curved outer and/or inner surfaces of the bulb, such as those that are configured to retain a cylindrical shaped beverage container, such as a can, bottle, cup and the like. The outer shape may be curved around the cylindrical shaped cup holder to produce a substantially uniform thickness ring or cylinder. Note that the cylinder may be incomplete as and end cup holder may produce an opening to the cup holder.

Seam: As used in this disclosure, a seam is a joining of: 1) a first textile to a second textile; 2) a first sheeting to a second sheeting; or, 3) a first textile to a first sheeting. Potential methods to form seams include, but are not limited to, a sewn seam, a heat bonded seam, an ultrasonically bonded seam, or a seam formed using an adhesive.

Sewn Seam: As used in this disclosure, a sewn seam is a method of attaching two or more layers of textile, leather, or other material through the use of a thread, a yarn, or a cord that is repeatedly inserted and looped through the two or more layers of textile, leather, or other material.

Slot: As used in this disclosure, a slot is a long narrow groove or aperture that is formed in an object.

Strap: As used in this disclosure a strap is a strip of leather, cloth, or other flexible material, often with a buckle, that is used to fasten, secure, carry, or hold onto something.

Strip: As used in this disclosure, the term describes a long and narrow object of uniform thickness that appears thin relative to the length of the object. Strips are often rectangular in shape.

With respect to the above description, it is to be realized that the optimum dimensional relationship for the various components of the personal floatation system described above and in FIGS. 1 through 9 include variations in size, materials, shape, form, function, and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by the personal floatation system.

It will be apparent to those skilled in the art that various modifications, combinations and variations can be made in the present invention without departing from the scope of the invention. Specific embodiments, features and elements described herein may be modified, and/or combined in any suitable manner. Thus, it is intended that the present invention cover the modifications, combinations and variations of this invention provided they come within the scope of the appended claims and their equivalents.

What is claimed is:

1. A personal floatation system comprising:

- a) a horseshoe shaped personal floatation device that is buoyant and comprises:
 - i) a closed end;
 - ii) a right armrest extending from a first side of the closed end to a right armrest free end;
 - iii) a cylindrical shaped right arm bulb configured on the right armrest free end;

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iv) a left armrest extending from a second side of the closed end to a left arm free end;

v) a cylindrical shaped left arm bulb configured on the left armrest free end;

vi) a cup cylinder configured in at least one of the left arm bulb or right arm bulb and having an opening extending into said armrest bulb from a top surface of said left arm bulb or right arm bulb;

vii) an end cup holder configured in at least one of the left arm bulb or right arm bulb and having an opening extending into said armrest bulb from the free end;

viii) an inner perimeter extending along the closed end and along the left armrest and right armrest; and

wherein the closed end extends in an arc from the right armrest to the left armrest; and wherein the personal floatation device is an inflatable personal floatation device.

2. The personal floatation system of claim 1, further comprising a detachably attachable cover configured to extend over and encase the personal floatation device.

3. The personal floatation system of claim 2, wherein the personal floatation device strap system comprises:

- i) a left extending strap that extends around the free end of the left armrest; and
- ii) a right extending strap that extends around the free end of the right armrest;

wherein the left extending strap and the right extending strap are configured to be secured to each other to couple the left arm to the right arm.

4. The personal floatation system of claim 2, further comprising:

- b) a personal floatation device strap system comprising:
 - i) a personal floatation device extending strap that extends around the free end of the left armrest or the right armrest;

wherein the personal floatation device extending strap is configured to extend to and be secured to the opposing left arm or right armrest to couple the left armrest to the right armrest by said personal floatation device extending strap.

5. The personal floatation system of claim 3, wherein the personal floatation device strap system further comprises an over-strap extending around the horseshoe shaped personal floatation device that secures the extending strap to the personal floatation device.

6. The personal floatation system of claim 5, wherein the personal floatation device further comprises:

- i) a first right armrest groove extending down from a top surface of the right armrest and substantially orthogonal to an extension direction of the right armrest;
- ii) a first left armrest groove extending down from a top surface of the left armrest and substantially orthogonal to an extension direction of the left armrest.

7. The personal floatation system of claim 6, wherein the over-strap extends around the first right armrest groove or the first left armrest groove to secure the extending strap to the personal floatation device.

8. The personal floatation system of claim 6, wherein the personal floatation device further comprises:

- i) a second right armrest groove configured more proximal to the right arm bulb than the first right armrest groove and extending down from a top surface of the right armrest and substantially orthogonal to an extension direction of the right armrest;
- ii) a second left armrest groove configured more proximal to the left arm bulb than the first left armrest groove and

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extending down from a top surface of the left armrest and substantially orthogonal to an extension direction of the left armrest.

9. The personal floatation system of claim 8, wherein the over-strap extends around the second right armrest groove or the second left armrest groove to secure the extending strap to the personal floatation device.

10. The personal floatation system of claim 1, wherein the personal floatation device further comprises:

- i) a first right armrest groove extending down from a top surface of the right armrest and substantially orthogonal to an extension direction of the right armrest;
- ii) a first left armrest groove extending down from a top surface of the left armrest and substantially orthogonal to an extension direction of the left armrest;
- iii) a second right armrest groove configured more proximal to the right arm bulb than the first right armrest groove and extending down from a top surface of the right armrest and substantially orthogonal to an extension direction of the right armrest;
- iv) a second left armrest groove configured more proximal to the left arm bulb than the first left armrest groove and extending down from a top surface of the left armrest and substantially orthogonal to an extension direction of the left armrest.

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11. The personal floatation system of claim 1, further comprising a neck extension that extends up from closed end.

12. The personal floatation system of claim 11, wherein the neck extension comprises a chin depression.

13. The personal floatation system of claim 12, wherein the chin depression is concave depression in the neck extension.

14. The personal floatation system of claim 1, further comprising strap-loops configured on an outside of each of the right armrest and left armrest.

15. The personal floatation system of claim 14, further comprising a strap-loop configured on an outside of the closed end.

16. The personal floatation system of claim 1, further comprising a detachably attachable cover configured to extend over and encase the personal floatation device and wherein the cover comprises strap-loops configured to extend from of each of the right armrest and left armrest when the cover is configured over the personal floatation device.

17. The personal floatation system of claim 16, wherein the cover further comprising a strap-loop configured to extend from of an outside of the closed end armrest when the cover is configured over the personal floatation device.

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